Analysing all the factors behind the long waiting times and optimizing it by providing them the insights from the predictions we get to improve the efficiency of Emergency Departments.

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Declaration

I declare that the work presented in this dissertation is, to the best of my knowledge and belief, original and my own work. The material has not been submitted, either in whole or in part, for a degree at this, or any other university.

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Abstract

Emergency Departments are vital for providing immediate care to acutely ill or injured patients. However, the increasing influx of patients has led to significant overcrowding and extended waiting times, which negatively impact patient satisfaction and outcomes. This study is about the analyses of a dataset with patient records from a UK NHS hospital's ED. By applying the statistical and machine learning techniques, the research identifies key factors affecting the ED efficiency and develops predictive models to assist operational staff in managing patient flow. This study helps to address resource allocation challenges by proposing operational strategies to optimize capacity of the hospital, minimize waiting times, and improve healthcare services of hospital in the emergency department.

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Chapter 1

Introduction

Emergency departments (EDs) are intricate systems that are required to operate continuously round the clock, for treating the patients coming in with emergency issues. Due to these reasons the ED's face Staffing issues, budget shortfalls, and a variety of other challenges on daily basis. Despite its importance, many of the issues faced by these departments remain unresolved, prompting researchers to explore innovative methods, such as computer simulations, to test and improve the efficiency of these complex systems. One of the most pressing issues in EDs, particularly in the United Kingdom, is the management of waiting times as they have a "4- hour Target" to see the patient within that time. These targets are a central aspect of hospital management efforts, as they always try maintaining that balance between quality and efficiency in the services they provide to the patients. Accurate forecasting models are essential for getting insights on this which will help in planning the resources accordingly and manage the emergency department efficiently.

The patient flow in the Emergency departments fluctuates with time, which makes it more complex for the management to handle. As this is a continuous process with a constant fluctuation in the patient flow, it is very important for the hospital management to be prepared for the least and highest number of patients possible coming in always to provide good services which is only possible when they have enough resources allocated always for the patients. Traditional methods of handling the waiting times, which often rely on historical data, are inadequate as the ED operations are dynamic nature. Recently, there have been a lot of statistical methods and Machine learning algorithms implemented for improving the accuracy of the waiting time predictions in the ED by offering a more reliable approach to forecast patient expectations and enhance their overall experience in the ED with enough resources (Pak et al. 2021).

As we know that the hospitals are key healthcare service provider, especially the emergencies services are the most important section of the hospital that provides treatment to very critical conditions which is a matter of life and death most of the times. Hence, performance of an ED has a significant impact on the patients, as it is responsible for the initial check, decision making for the next steps, stabilizing the patient, then treatment for all the conditions and then the recovery. Planning and organising the ED resources efficiently is one of the major factors that will affect the performance towards the patients. One of the main goals of hospital management is to increase patient satisfaction while reducing operational costs, which includes minimizing waiting times in the ED.

Chapter 2

Literature Review:

Emergency departments are critical components of the healthcare system, offering urgent medical care to patients with a wide range of health conditions. However, one of the most persistent challenges faced by EDs all over the world is to manage the patient's waiting times. Prolonged waiting times can lead to overcrowding, patient dissatisfaction, and potentially adverse health outcomes. Various studies have been conducted to explore different factors influencing waiting times, including patient characteristics, triage processes, and resource allocation. This literature review synthesizes key findings from few significant studies that have contributed to our understanding of the patterns related to waiting times in ED, while focusing on patient types, triage processes, simulation-based optimization methods, regression models, and the specific challenges faced by patients with psychiatric illnesses or social issues.

2.1. Patient Characteristics, Triage processes and their impacts on the Waiting Times

In the study titled **"Which patients wait longer to be seen and when?"** (Elkum et al. 2009), a comprehensive statistical approach was utilized to explore waiting times in a tertiary care emergency department (ED) setting. The research incorporated both **bivariate** and **multivariate** analysis methods to determine how selected patient characteristics, including age, day of arrival, time of arrival, and triage category, influenced the waiting time between triage and being seen by a physician.

The study analysed **2,187 patient records**, collected over four months, using a **retrospective study design**. Each patient record included timestamps for registration, triage, and physician consultation, which were recorded by registration clerks, triage nurses, and evaluating physicians. The triage levels were classified into three categories: **emergent**, **urgent**, and **non-urgent**, reflecting the severity of the patient's conditions.

Analytical Techniques like Bivariate Analysis & **Multivariate Analysis**: The researchers used non-parametric statistical tests to explore associations between waiting time and independent variables like age, sex, day of arrival, and triage category.

Kruskal-Wallis tests were employed to compare waiting times across different groups, as the waiting time data was highly skewed. **Weibull regression analysis** was used to model the relationship between the dependent variable (waiting time) and multiple independent variables (such as age, day of arrival, and triage category).

This regression model was particularly useful due to the **positive skewness** in the waiting time data, which made conventional linear models unsuitable. The Weibull model is adept at handling skewed distributions, allowing for a more accurate assessment of how multiple factors influenced waiting times.

They found that **age** was a significant predictor of waiting time, with older patients waiting longer than younger ones. Children (0-14 years) experienced the shortest waits, while patients over 65 had significantly longer waiting times.

Triage Category also significantly affected waiting time. Interestingly, patients in the **urgent** category often faced longer waits than those in the **emergent** or **non-urgent** categories. This was attributed to variability and inconsistencies in triage classification.

Day and Time of Arrival: Patients arriving on **Sundays and Wednesdays** experienced the longest waits. Additionally, the time of day also played a critical role, with patients arriving between **00:01 and 06:00 hours** or between **12:01 and 18:00 hours** experiencing significantly longer waiting times.

The results of this study shed light on the operational challenges faced by the ED, particularly in managing patient flow during peak times. By identifying factors that contribute to prolonged waiting times, such as age and day of arrival, the study suggests the need for more **standardized triage guidelines** to reduce variability in patient prioritization. Furthermore, the study emphasizes the importance of refining triage systems, such as implementing more detailed frameworks like the **Canadian Triage and Acuity Scale (CTAS)**. This system would potentially offer a more consistent approach to patient classification, minimizing the influence of non-clinical factors, such as subjective nurse judgments or administrative inefficiencies.

The study's use of **Weibull regression** to model waiting times highlights a robust approach to handling skewed data, offering a more nuanced understanding of how multiple factors concurrently affect ED operations. These findings can inform targeted interventions, especially for vulnerable populations like the elderly, helping to **optimize resource allocation** and reduce waiting times.

In conclusion, Elkum et al. (2009) provide a valuable foundation for improving ED efficiency by identifying critical factors influencing waiting times and proposing a structured, standardized approach to patient triage.

2.2. Simulation based analysis in Emergency departments:

Yousefi et al. (2020) presented a systematic review focusing on various simulation-based optimization techniques applied in emergency departments (EDs) to improve their operational efficiency. The review analysed 38 studies spanning from 2007 to 2019, focusing on methodologies such as Discrete

Event Simulation (DES), Agent-Based Simulation (ABS), and System Dynamics (SD). The study categorized objectives such as reducing patient length of stay (LOS), optimizing resource allocation, and decreasing patient waiting times. Various optimization methods were employed, including Genetic Algorithms (GA), Sequential Stochastic Comparison, and other heuristic approaches. A key finding was the growing utilization of multi-objective optimization frameworks that integrate multiple simulation techniques, providing better resource management solutions. The review emphasized the importance of choosing the right simulation technique depending on the specific ED challenge, whether it is patient overcrowding, staff scheduling, or budget constraints.

In another case study conducted by Sasanfar et al. (2021), a Discrete Event Simulation (DES) model was applied using Arena software to optimize patient waiting times and staff allocation in a hospital ED in Semnan Province, Iran. The simulation aimed to reduce waiting times for both internal and surgical emergency patients by redesigning the ED's layout and reallocating staff resources. The study involved three scenarios that assessed different layouts of the ED. The results demonstrated significant reductions in patient waiting times, with internal emergent patients experiencing a 23.18% reduction in waiting time and surgical emergent patients seeing an 81.7% reduction. This was achieved by placing critical areas like the revival room closer to the entrance and improving the overall workflow. The optimization was also extended to staff allocation, ensuring optimal use of nurses, residents, and general doctors in each ED ward.

Both studies underscore the importance of simulation-based approaches in enhancing ED operations by providing evidence-based strategies for layout redesign, resource allocation, and staff scheduling. They highlight the potential of using simulation tools to test various configurations and optimize key performance indicators, such as patient waiting times and staff utilization, in a risk-free environment.

2.3. Regression Models – Logistic, Ridge and Lasso Models for the waiting time predictions in ED

2.3.1. Ordinal Regression Model (Ataman et al. 2021)

The study on "Predicting Waiting and Treatment Times in Emergency Departments Using Ordinal Logistic Regression Models" applied an ordinal logistic regression (OLR) approach to classify emergency department (ED) patients based on their waiting and treatment times. Patients were categorized into predetermined ordered groups according to their waiting times (<10 minutes, 10–60 minutes, or ≥60 minutes) and treatment times (<10 minutes, 10–120 minutes, or ≥120 minutes). The OLR model was chosen for its suitability in handling ordered output variables such as waiting and treatment times, allowing the identification of the key factors influencing longer or shorter wait time.

The analysis included significant predictors such as patient age, mode of arrival, and ICD-10 diagnosis codes. It was found that older patients, those arriving by ambulance, and patients with specific ICD-10 diagnoses experienced longer treatment times and, in some cases, shorter waiting times. While the triage category had a significant effect on treatment time predictions, it did not predict waiting times. Interestingly, despite gender differences in both waiting and treatment times, gender was not a significant predictor in either model.

The performance of the OLR models was assessed using classification accuracy, which was calculated based on the number of correct predictions. The overall accuracy for predicting waiting times was 52.2%, while the accuracy for treatment time predictions was higher at 66.4%. The study emphasized that while gender differences were evident in univariate analyses, their predictive power was suppressed by other variables in the regression models.

2.3.2 Ridge and Lasso Regression Models (Pak et al. 2021)

The second study, "Patient Waiting Time Analysis Based on Patient Acuity Using Ridge and Lasso Regression," focused on regression techniques like Ridge and Lasso to predict waiting times based on patient acuity in EDs. These methods were particularly effective in addressing multicollinearity among predictors, which is common in healthcare data. By including predictors such as patient triage category, arrival time, and diurnal fluctuations, the models aimed to provide accurate estimates of waiting time.

Lasso and Ridge regression both performed significantly better than traditional methods such as rolling averages, particularly in reducing underprediction of waiting times—a critical concern in ED operations. Lasso showed a 21% reduction in mean squared prediction error (MSPE) compared to the rolling average, making it the most accurate model for waiting time prediction.

Quantile regression, random forest models, and ordinary least squares (OLS) regression were also tested in the study, but Lasso and Ridge consistently outperformed them in terms of MSPE and the percentage of underpredicted waiting times. The study further highlighted that including queueing and service flow variables was critical to improving the prediction accuracy. The importance of these predictors was reflected in the significantly better performance of models that included them.

In summary, both papers utilized advanced statistical models to predict ED waiting times, with the ordinal logistic regression focusing on classification accuracy across ordered groups, while the Ridge and Lasso regression methods offered robust predictions with lower error rates, making them highly suitable for real-time ED management.

2.4. Wating Time behaviours in NHS and NHS targets in the Emergency departments:

The study titled *"Waiting List Behaviour and the Consequences for NHS Targets"* by (Bowers 2010) provides a detailed examination of how patient waiting lists are managed within the NHS and how they affect the system's ability to meet its performance targets. The research focuses primarily on elective treatments, specifically in orthopaedics, using data from NHS Fife in Scotland between 2004 and 2007. During this period, the NHS was actively working towards reducing waiting times for elective care, aiming to meet the national target of treating patients within 18 weeks of referral.

The paper emphasizes that waiting list management in the NHS goes beyond simple "first-in-first-out" (FIFO) queuing disciplines, which are commonly assumed in many simulations of patient flow. Instead, waiting times within the NHS are influenced by various factors, including the clinical urgency of the patient's condition, capacity constraints, and even patient behaviour, such as their willingness to delay treatment. The research introduces a more complex model to capture these dynamics, particularly focusing on how different management strategies, such as outsourcing surgical procedures to the private sector or increasing recurrent capacity (e.g., additional clinics or theatre sessions), can help achieve waiting time targets.

From a technical perspective, Bowers develops a simulation model that incorporates the entire patient journey from referral to treatment, accounting for various stages, such as outpatient consultations, diagnostics, and surgery. This model is distinct in its incorporation of empirical waiting time distributions rather than relying on theoretical queuing models. These empirical distributions are used to simulate the likely outcomes of different management scenarios and assess their impact on overall waiting times. The model includes prioritization based on clinical need, recognizing that patients with higher urgency will receive faster treatment, while routine cases may experience longer waits.

One key observation of the study is the evolving distribution of waiting times over the period of analysis. The introduction of NHS targets led to changes in how patients were prioritized, with evidence of a bimodal distribution emerging in some cases—where urgent patients were treated quickly, and routine patients faced much longer waits. The study concludes that while the NHS was successful in reducing average waiting times, particularly the long tail of those waiting the longest, the focus on meeting specific targets sometimes led to uneven patient experiences, with resources being heavily allocated to avoid breaches of the 18-week maximum wait. The paper also discusses how variability in waiting times can stem from both systemic and patient-level factors. For instance, the availability of resources such as theatre time and diagnostic equipment can fluctuate due to staffing levels or seasonal demand changes, while patient behaviour, such as choosing to defer treatment, also plays a

significant role in shaping the overall waiting list dynamics. By addressing these factors through simulation, the study provides valuable insights into how NHS managers can better forecast waiting times and make informed decisions to meet national performance targets.

In sum, this paper contributes to the broader discourse on health service optimization by highlighting the importance of incorporating real-world complexities into simulation models of patient flow. It suggests that while increasing capacity and efficiency can help reduce waiting times, attention must also be paid to how patients are prioritized and how their behaviour affects overall system performance.

The paper titled *"Waiting Time Statistics in Britain: A Critical Review"* (Godden and Pollock 2019) evaluates the methodology behind the collection and reporting of waiting time statistics within the NHS. The study focuses on inconsistencies and limitations in the way waiting times are measured across different regions, namely England, Scotland, and Wales. The authors argue that these inconsistencies often lead to an inaccurate picture of patient access to care, which can ultimately undermine the effectiveness of NHS performance targets.

One of the central point's made in the paper is the variability in data collection methods. In England, waiting time statistics tend to focus on meeting national targets, with hospitals being incentivized to report favourable results. However, these statistics often exclude certain categories of patients, such as those who miss appointments or voluntarily defer treatment, thereby skewing the overall picture. In contrast, the authors suggest that more comprehensive data collection methods that track the entire patient journey, from referral to treatment, would provide a more accurate reflection of the NHS's performance.

The technical aspects of the study revolve around the analysis of the statistical methods used to calculate waiting times. The authors highlight several methodological issues, including the exclusion of long-waiting patients from official statistics and the emphasis on average waiting times, which can mask the experiences of those who wait the longest. The study suggests that relying on average waiting times can lead to complacency in addressing the needs of patients at the extreme end of the distribution, who often experience the longest delays in receiving care.

To address these issues, the authors advocate for a more nuanced approach to measuring waiting times, one that includes patient-level data and considers the entire duration of the patient's interaction with the healthcare system. This would involve tracking patients from their initial referral, through diagnostic procedures, to their eventual treatment. Such an approach would not only provide a clearer

picture of waiting times but also help identify specific bottlenecks in the system, such as delays in accessing diagnostics or specialist consultations.

The paper also touches on the political and managerial implications of the current waiting time statistics. The emphasis on meeting targets, particularly in England, can lead to unintended consequences, such as hospitals prioritizing patients who are nearing the target threshold while neglecting those who are less urgent but still in need of care. This target-driven approach can create perverse incentives, where the focus shifts from improving patient care to simply meeting numerical targets. The authors call for a fundamental rethink of how waiting time statistics are used to measure NHS performance. They argue that a more comprehensive and patient-centred approach to data collection would enable healthcare providers to allocate resources more effectively and ensure that patients are treated based on clinical need rather than the pressures of meeting arbitrary targets.

In conclusion, this paper underscores the importance of developing more reliable and accurate methods for collecting and reporting waiting time statistics in the NHS. By addressing the methodological flaws in the current system, the authors believe that healthcare providers can better understand the true extent of waiting times and take more effective steps to reduce them.

2.5. Challenges in Managing Patients with Psychiatric Illnesses

The paper *"Patients with Psychiatric Illness: Length of Stay in the Emergency Department"* (Weiss et al. 2012) addresses the unique challenges faced by emergency departments (EDs) when managing psychiatric patients, focusing specifically on the factors that contribute to their extended length of stay (LOS). Psychiatric patients often experience significantly longer ED stays compared to patients with other conditions, primarily due to the need for specialized psychiatric evaluations and the limited availability of psychiatric beds. The study uses a mixed-effects regression model to assess various factors impacting LOS, including demographic details, clinical characteristics, and specific hospital processes. Notably, the presence of alcohol in patients' systems, the need for diagnostic imaging, and the use of physical restraints were all identified as key contributors to extended ED stays. The study found that psychiatric patients, on average, remained in the ED for 11.5 hours, with certain subgroups such as those awaiting psychiatric beds or requiring additional diagnostics experiencing significantly longer stays.

One of the critical insights from the study is the impact of systemic bottlenecks, such as the availability of psychiatric beds, which often result in prolonged stays even after a decision to admit has been made. Delays in patient transfers, particularly for those needing specialized psychiatric care outside the ED, were a major factor contributing to extended LOS. Additionally, the paper highlights how non-clinical factors like patient intoxication further exacerbate these delays, as psychiatric assessments are often

deferred until the patient is sober. The study advocates for the establishment of more integrated care pathways, including the creation of dedicated psychiatric observation units within EDs to expedite care for psychiatric patients. By improving coordination between EDs and psychiatric units, and by streamlining processes such as diagnostic imaging, the overall burden on EDs can be reduced, leading to shorter waiting times and better patient outcomes. This study emphasizes the need for targeted interventions and the development of more efficient systems to manage psychiatric cases in ED settings.

2.6. Stochastic Models for ED Waiting Time Analysis

The paper titled *"A Stochastic Model for Waiting Time Analysis in Emergency Departments"* (Daldoul et al. 2018) offers a detailed mathematical approach to understanding and predicting waiting times in emergency departments (EDs), emphasizing the role of randomness and uncertainty in patient arrivals and service times. The authors propose a stochastic mixed-integer programming model specifically designed to capture the variability in ED operations, which traditional deterministic models often overlook. The study focuses on La-Rabta, a public hospital in Tunisia, where the unpredictability of patient demand often causes significant delays in care delivery. By modelling these unpredictable factors, the authors aim to provide a more accurate representation of the complex dynamics within an ED, such as fluctuating patient volumes and varying service times. The stochastic model is structured to optimize the allocation of human resources (including physicians and nurses) and material resources (such as available beds) to minimize patient waiting times while accounting for these random variables.

One of the key aspects of the model is its ability to handle multiple patient queues simultaneously, which reflect different stages of patient care within the ED. The study identifies six major queues: triage, general and surgical assessments, auxiliary exams (such as X-rays and laboratory tests), life-threatening emergencies, and bed assignments for admitted patients. Each queue is modelled with varying levels of priority and resource demands, allowing the stochastic model to optimize the distribution of resources across different stages of patient care. This complex setup ensures that the model addresses not only the randomness in patient arrivals but also the prioritization of life-threatening cases, which must be handled immediately, over less urgent medical needs. The technical core of the study is the application of the Sample Average Approximation (SAA) method, a Monte Carlo-based approach that is well-suited for solving stochastic optimization problems. SAA works by generating many patient arrival scenarios based on historical data, simulating various random outcomes, and then calculating the average results to approximate the best resource allocation strategies. The stochastic model uses these approximations to determine how many resources should be allocated to each queue during different periods of the day, ensuring that resources are available when patient demand is highest. By running this model on real-world data from La-Rabta hospital, the

researchers were able to demonstrate a 23.24% reduction in average waiting times. This significant improvement was largely achieved by reallocating resources during peak hours, ensuring that more physicians and nurses were available when patient arrivals surged.

It highlights the significant advantages of using stochastic models in ED management by accounting for the inherent variability in patient arrivals and service times, unlike traditional deterministic models. This approach allows healthcare administrators to better manage fluctuations in demand, leading to more efficient resource utilization and improved patient flow. By incorporating stochastic elements, ED operations can dynamically allocate resources, prioritize urgent cases, and minimize waiting times. The use of the Sample Average Approximation (SAA) method enhances the model's accuracy, enabling simulations of different patient scenarios to optimize resource allocation. Overall, this approach offers a valuable tool for reducing overcrowding and improving patient outcomes in unpredictable, high-demand environments.

2.7. Markov Models and Priority Queuing in Emergency Departments

The paper"Using a Priority Queuing Approach and Markov Models to Reduce ED Waiting Times" (Hou and Zhao 2020) explores advanced mathematical techniques to optimize patient flow in emergency departments (EDs). The research integrates priority queuing theory with Markov models to address the problem of patient overcrowding and long waiting times in EDs. The authors designed a multi-class, multi-server system where patients are prioritized based on the severity of their condition. In the model, high-priority patients, such as those requiring resuscitation or critical care, receive immediate attention, while patients with less severe conditions wait longer. The queuing system is non-pre-emptive, meaning that once a physician begins treatment, the patient must be attended to without interruption, even if higher-priority cases arise. The study uses Markov queues to predict patient arrival and service times and provides an approximation formula for more general queuing systems, offering a practical tool for hospital administrators. Simulation experiments demonstrate the accuracy of the models, with significant improvements in waiting times for high-priority patients.

Furthermore, the research highlights the computational efficiency of the proposed method, which avoids the complexity of iterative algorithms often required by traditional models. This is particularly important in real-world ED operations, where patient arrival rates and treatment times vary. The paper also includes a detailed case study from a Chinese hospital, showing that the queuing model effectively reduced waiting times for critical patients while maintaining service quality for lower-priority patients. The authors argue that the system can be easily adapted for use in other hospitals and ED settings, offering a scalable solution to the growing problem of ED overcrowding. Through numerical analysis and simulation, the study demonstrates that priority queuing, combined with Markov modelling, can

serve as a robust decision-making tool for healthcare administrators seeking to optimize resource allocation and improve patient outcomes.

2.8. Prediction Tool for ED Waiting Times

"A Two-Step Predictive Model for Early Detection of Emergency Department Patients with Prolonged Stay and its Management Implications"(d'Etienne et al. 2021) focuses on developing a predictive tool designed to identify ED patients likely to have extended lengths of stay. Using data from over 110,000 patient visits, the researchers created a two-step model aimed at improving ED efficiency. The first step predicts the likelihood of a patient requiring certain hospital resources (e.g., intravenous therapy, lab tests, and imaging services) during their ED visit. The second step builds on this prediction by estimating whether a patient's stay will exceed 16 hours, factoring in both the anticipated resource use and real-time ED crowding levels. The model's overall accuracy was 67.8%, with the C-index ranging from 0.72 to 0.82, indicating strong predictive power. The simulation-based intervention showed that by moving patients predicted to have prolonged stays to a Clinical Decision Unit (CDU), the overcrowding status in the ED was reduced by 12%, saving over 129 bed hours daily. The study underscores the value of this two-step predictive model in helping ED administrators anticipate patient needs and make informed decisions about resource allocation. The model's ability to predict prolonged stays early in a patient's visit allows for timely interventions, such as transferring patients to a CDU, thereby optimizing bed usage and improving patient flow. The application of discrete event simulation (DES) further enhanced the model by enabling the simulation of different patient arrival patterns and testing various intervention scenarios. This novel predictive tool offers ED managers a data-driven approach to managing crowding and improving the overall quality of care in high-demand settings.

The studies reviewed in this literature review provide a comprehensive understanding of the various factors influencing waiting times in EDs, including patient characteristics, triage processes, simulation-based optimization methods, regression models, and the specific challenges faced by patients with psychiatric illnesses. The addition of recent studies on prediction tools and critical reviews of waiting time statistics further enriches the discussion, highlighting the ongoing efforts to improve ED efficiency and patient satisfaction. Together, these studies underscore the complexity of managing ED operations and the need for a multifaceted approach to reducing waiting times and improving patient outcomes. By integrating the findings from these studies, healthcare providers can develop more effective strategies to address the challenges of ED overcrowding and enhance the quality of care provided to patients.

Chapter 3

Exploratory Data Analysis:

The Dataset used in this study consists of 610,439 patient records from April 2019 to June 2024, providing a comprehensive view of the activities in the Emergency Department and the patient flow over the years. It includes both numerical and categorical variables like:

1. **ED A&E Number**: A unique identifier for each emergency department attendance.
2. **ED Patient ID**: A unique identifier for each patient.
3. **ED Admitted:** A categorical variable indicating whether the patient was admitted to the hospital following their ED visit.
4. **ED 4-Hour Breach:** A categorical variable that indicates whether the patient's wait time exceeded the four-hour standard.
5. **ED Arrival Date and Time**: Timestamp indicating when the patient arrived at the ED.
6. **ED Minutes from Arrival to Triage/Seen**: The time in minutes from arrival to being triaged or seen by a healthcare professional.
7. **ED Departure Date and Time**: Timestamp marking when the patient left the ED.
8. **ED Attendance Category Code**: A code representing the category of attendance.
9. **ED Presenting Complaint and Primary Diagnosis**: The main complaint and diagnosis were recorded for the visit.
10. **Age**: The patient's age at the time of the ED visit.
11. **Arrival Mode**: The mode of transportation used to arrive at the ED.
12. **Arrival to Departure Period (Minutes)**: Total time spent in the ED, calculated from arrival to departure.
13. **Sex**: The gender of the patient.

To gain more insights on the data, exploratory data analysis was conducted which is structured into different steps focusing on various aspects or variables of the dataset.

Count graphs

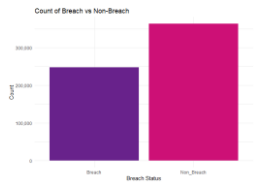

Fig.1 Breach Vs Non-Breach

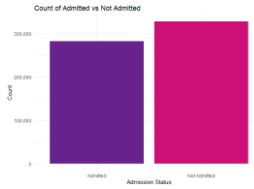

Fig.2 Admit Vs Non-Admit

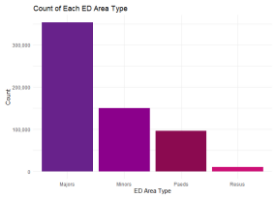

Fig.3 ED Area

Breach status from the **Fig 1** which compares the number of patients who breached the "4-hour target time" and it demonstrates that most of the patients did not breach the standard waiting time which means that the hospital is generally able to manage the emergency department and maintain the expected waiting time standards, though there as an scope of improvement indicated as there still breaches shown may be during the peak times or high acuity cases.

Admission status from the **Fig 2** compares the number of patients who are getting admitted and leaving the hospital after getting treated on same day. This graph shows that a significant number of patients are not getting admitted. The "Not Admitted" count exceeding the "Admitted" count suggests that the hospital management is able to treat a large portion of patients on the same day without any admission requirement. This helps in reducing the in-patients with an effective triage and treatment protocols but might be affecting the resources required for the existing in-patients as they are busy with the out-patients.

ED Area in **Fig 3** shows us the distribution of the patients from different categories, "Majors" are the highest followed by "Minors", "Paediatrics" and "Resus". The critical cases are less but high volume of "Majors" must be noted to get the resource allocation done precisely.

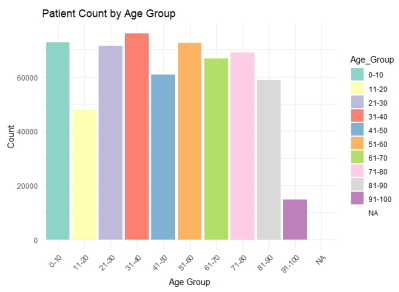

Fig.4 Age Distribution

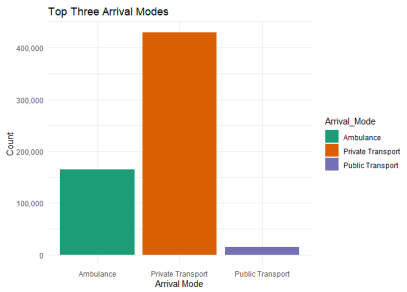

Fig.5 Arrival Mode

Age Distribution in **fig 4** shows that the 31-40 and 41-50 age groups are the most represented in the ED, followed by the 51-60 age group. Younger children (0-10 years) also have a significant presence, while the elderly (over 70 years) have lower counts, which could be due to alternative

care pathways or lower visit rates. This distribution highlights the need for ED resources to cater to middle-aged adults and paediatric cases predominantly.

“Arrival Mode” in **fig 5** shows that most patients arrived via private transport, followed by ambulance and public transport. The dominance of private transport could be influenced by various factors, including the availability of ambulances, the distance from the hospital, and the urgency of the condition.

This information is crucial for planning emergency service resources and ensuring that the ED is equipped to handle the high volume of privately transported patients. Given that reducing the waiting time i.e. breaches in the emergency is the focus of this study we delved deeper into the Breached and Not Breached patients to look at the patterns and identify the factors leading to the breaches.

Patients who breached the 4-hour limit:

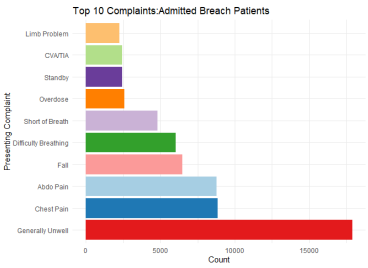

Fig 6 Top 10 Admit Breach Complaints

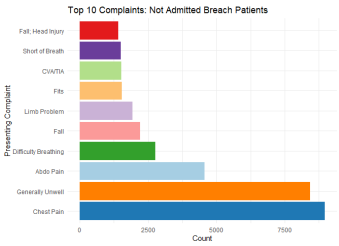

Fig 7 Top 10 Not Admit Breach Complaints

The graph in **Fig 6 and 7** visualizes the top 10 complaints that are getting breached for both “Admitted” and “Not Admitted”. Among the admitted patients who breached, the most common complaints were "Generally Unwell," "Chest Pain," and "Abdominal Pain,". These complaints are often associated with conditions that require comprehensive medical evaluation and suggest that patients with more severe or complex conditions are more likely to be admitted, even if it results in breaching the “4-hour” standard.

In contrast, the not admitted patients who breached show slightly a different distribution but “Generally Unwell” and “Chest Pain” remain the top two complaints. However, issues like “Falls” and “Limb Problems” are generally not critical and need less intensive treatment though these patients are experiencing delays but managed to be treated adequately without any need for admission in the hospital.

This can be considered that the breaches are not only because of the inefficiency or lack of resources in the hospital but also depend on the complexity or severity of the issues patients come in daily.

Arrivals of Breached patients both Admitted and Not Admitted by hour of the day:

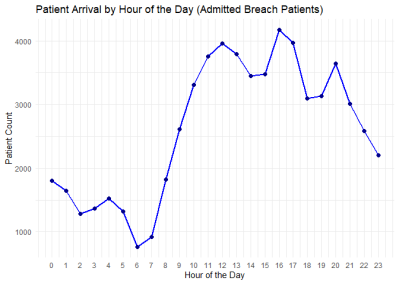

Fig.8 Admit-Breach Arrivals by Hour

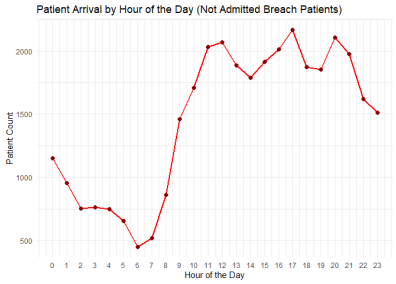

Fig.9 Non-Admit -Breach Arrivals by Hour

Both the graphs in **Fig 8 and 9** show us the arrival of the patients who are breached both under the "Admitted & Not Admitted" Categories. It's evident that the patterns exhibited here are quite similar with peaks around 10 AM, and 1 PM troughs around early morning and evening which means that the peak hours are from late afternoon to evening for both the category patients. Hence, these points highlight the factors that must be considered like staffing and other required resource allocations to handle the influx effectively.

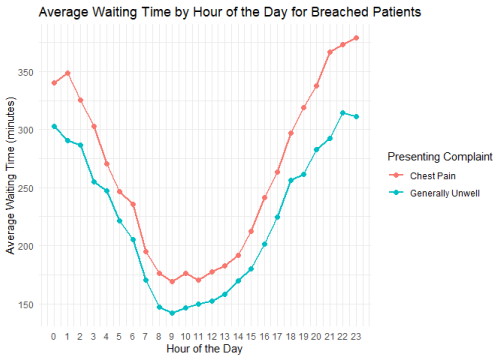

Fig.10 Average Waiting Time by Hour of the day for Breached Patients.

The Average Waiting time in the **Fig 10** shows the top two issues patients coming in with "Chest Pain" and "Generally Unwell" fluctuating throughout the day and can see a peak in the late evening from around 5PM to 2AM. This correlates with the patient arrivals that were observed earlier where patients come in during the daytime from 10 AM to 1PM when the staff experiences the high influx of patients which increases the workload on the available resources and as this continues the demand for medical attention exceeds the resources and facilities available, leading to the longer wating times.

However, this is fine with less critical patients, but more concerning would be for the ones who are very critical as this will be problematic for them to wait for longer time. Hence, it is worth noting that the cumulative pressure increasing in the ED as the day progresses should be observed and taken care of by allocating the resources and required facilities during these peak hours to minimize the breaches in the hospital.

Patients who did not breach the 4–hour limit:

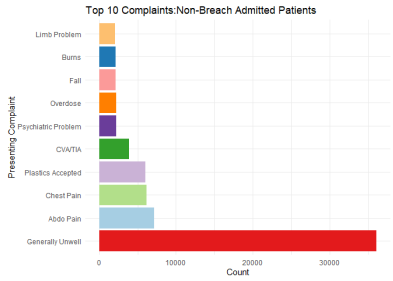

Fig.11 Top 10 Complaints Admit-Non-Breach

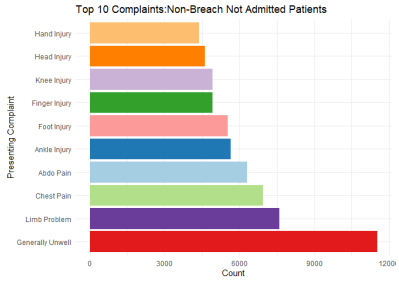

Fig.12 Top 10 Complaints Non-Admit-Non-Breach

The graphs in **Fig 11 and 12** illustrate the top 10 "Presenting Complaints" for the "non-Breach" patients who were either "admitted or not admitted". In both cases "Generally Unwell" is the most frequent complaint. "Admitted" category patients have been seen and immediately admitted on time and "Not admitted" patients mostly with injuries that can be treated without the need for admission into the hospital were also handled well within the "4-hour window".

Patient Arrivals by hour of the day for Non-Breach Patients:

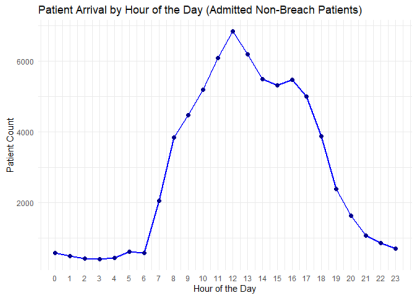

Fig.13 Admit-Non-Breach Arrivals by Hour

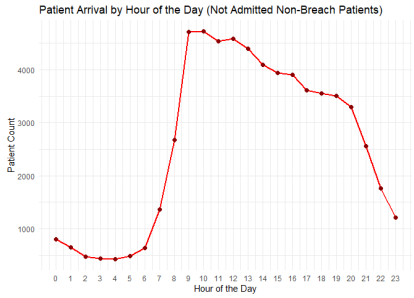

Fig.14 Non-Admit-Non-Breach Arrivals by Hour

The patient arrival patterns for non-breach admitted and not admitted patients in **Fig 13 and 14** show a significant influx during the late morning, peaking around 11 AM to 1 PM, and then gradually declining through the afternoon and evening. The similarity in these patterns across both categories suggests that despite the high volume of patients during these hours, the hospital was able to manage the patient flow effectively, preventing breaches. This indicates that the resources and staff allocation during these peak hours were sufficient to handle the patient load.

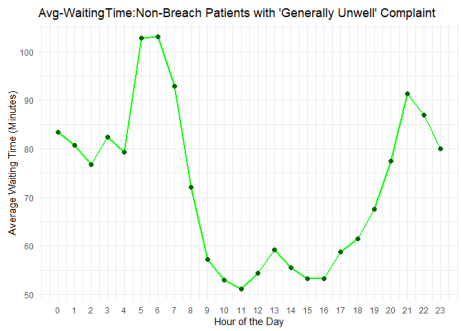

Fig.15 Average Waiting "Generally Unwell" Non–Breach Patients

The average waiting time graph in **fig 15** for non-breach patients presenting with "Generally Unwell" complaints shows fluctuations throughout the day, with a noticeable peak in the early morning around 6 AM and late evening around 9 PM. These times coincide with lower patient volumes, suggesting that even with fewer arrivals, certain patients still experienced longer waits, possibly due to a reduced staffing level during these off-peak hours. However, the overall shorter waiting times during peak hours indicate that the ED was able to manage the high influx of patients effectively, ensuring that those with serious complaints were attended to promptly and within the "4-hour target".

Weekdays Vs Weekend Patterns:

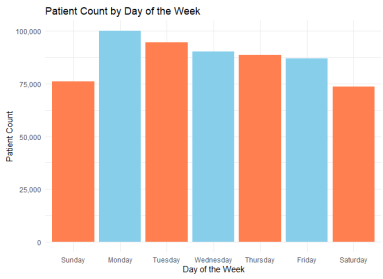

Fig.16 Patient by Days of the Week

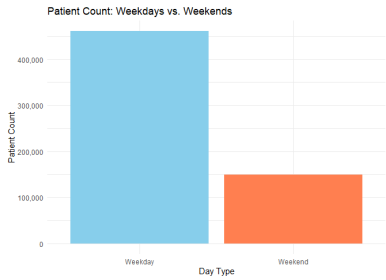

Fig.17 Patient Count Weekdays Vs Weekends

Moving forward, we examined the patient arrival patterns during weekdays and weekends and how it varies by hours of the day.

The graph in the **fig 16** shows distribution of patient arrivals across the days of the week. Monday consistently shows the highest patient count, followed closely by Tuesday and Wednesday. The rest of the week sees a slight decline, with the lowest counts on Saturday. This trend indicates that the start of the week is particularly busy for the ED, possibly due to patients delaying visits over the weekend or seeking care after the weekend's limited healthcare access.

The graph in the **fig 17** provides a clear comparison between weekdays and weekends, showing that weekday patient volumes are substantially higher. The total patient count on weekdays far exceeds that of weekends, reinforcing the observation that the ED is under greater strain during

the work week. This disparity highlights the need for strategic staffing and resource allocation to accommodate the increased demand during weekdays, particularly on Mondays and mid-week

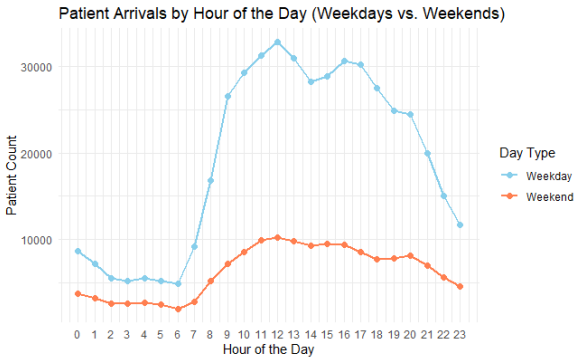

Fig.18 Patient arrivals by hour of day Weekday Vs Weekends

As evident from the Weekday vs. Weekend comparison graphs, patient arrivals are significantly lower on weekends compared to weekdays in **fig 18**. This trend is consistent in the subsequent graph, which also shows that the peak arrival times remain the same, with the highest influx occurring between 10 AM and 7 PM, aligning with the patterns observed in the previous patient arrival patterns. These suggests that the flow is very high and challenging during the weekdays than weekends.

Breaches by Days of the Week, Hour of the day and ED-Area:

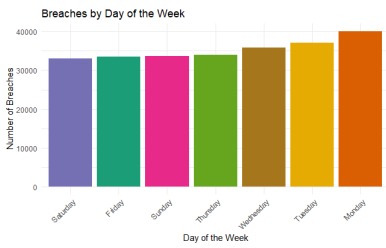

Fig.19 Breaches by Days of Week

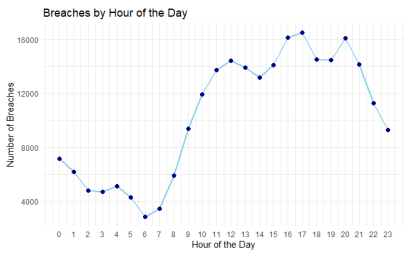

Fig.20 Breaches by hour of the Day

As the patient flow is higher on “Mondays” as observed above the breaches on Mondays are highest in **fig 19**, as the patients delay or neglect the visits during the “weekends”.

The Patient arrival observations from the previous sections reflect here showing the breaches during the 4PM – 8PM which again proves that the patients coming from around 10AM are gradually increasing by evening and the capacity of the ED exceeds which leads to the breaches due to lack of resources again. However, during the morning hours, the breaches are comparatively less due to low patient volumes on average for the whole week in **Fig 20**.

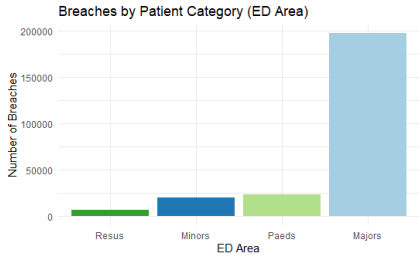

Fig. 21Breaches by Patient Category in ED Area

Fig 21 shows that majority of the breaches occur in the "Majors" category, indicating that these high-complexity cases are a significant factor in delays, contributing far more to breaches compared to the other three categories, which have considerably fewer breaches.

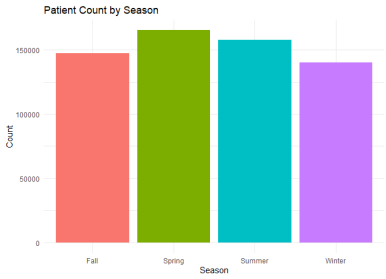

Fig.22 Patient Count by Season

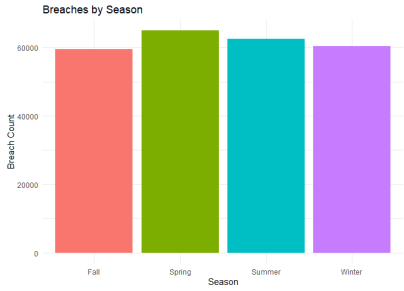

Fig.23 Breaches by Season

The Seasonal analysis in **fig 22 and 23** shows that patient arrivals and breaches peak in Spring and Summer, with lower volumes in Fall and Winter. The higher patient numbers during these seasons lead to more breaches, indicating the added strain on ED resources.

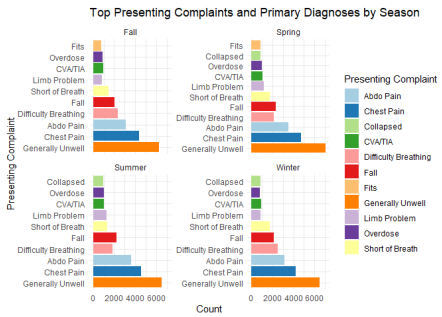

Fig. 24 Top Complaints by Season

The analysis of presenting complaints across seasons in **Fig 24** re confirms that "Generally Unwell" remains the most common complaint, consistent with previous findings.

In summary, from the exploratory data analysis we got our key variables that will be used in our further analysis where we will delve deeper into the reasons behind the long waiting time focusing on the ED Area types, Breaches, Admissions and Patient Arrivals by hour of the day as they seem to give us some significant insights about the waiting times.

Chapter 4

Methodology:

Exploratory Analysis provided a detailed overview of the patient flow and the activities in the emergency departments which includes all categorical and numerical variables with key variables as “ED 4 hr Breach”, “ED Arrival date and time”, “ED Presenting Complaint”, “ED Admission status”, “Arrival Mode”, “ED Area”, “Age”, etc.,

From the insights gained from EDA, I decided to Extend the analysis further to understand and evaluate the potential impact of the COVID-19 pandemic on the patient flow in the emergency department.

Pre-Covid and Post-Covid impact on the Patient flow in the Emergency Department:

The data set was divided into 2 subsets with a cut-off date of March 11, 2020 as this is the date declared by the World Health Organization for the pandemic. This helped in the comparative analysis between both before and after periods of the global health crisis.

Step 1: Segregating the data

Pre-COVID data subset of the data includes the patient's records from April 2019 to March 10, 2020 which gives us the normal conditions of the hospital operations before the pandemic in the emergency department.

The post-COVID data subset of the data includes the records of the patients from 11 March 2020 to June 2024 which is the after covid period which reflects the potential disruptions or changes in the patient flow and hospital operations due to the lockdown and the recovery phase in the healthcare sector.

Step 2: Aggregating the Data into a Monthly pattern

After splitting the data into 2 subsets, I have decided to check the monthly aggregate to see the pattern and analyse the trend in the patient flow over the period in both durations as this would be a very crucial point to consider if there are any significant deviations in the flow of the patients that might be caused by the pandemic situation world-wide including the recovery phase.

Step 3: Visualising the Patterns by plotting the graphs

A line graph was plotted to visually represent the trend over time in the patient flow which will highlight the abrupt changes due to the pandemic and the later recoveries in the flow.

After visualising the patient flow in the line graph plotted over the years, I decided to check and highlight the outliers or drastic shifts in the patient volumes during both periods by generating a box plot.

Step 4: Statistical test to compare both the periods

Welch Two Sample t-test is a variation of the traditional t-test that is used when we are comparing the means of two independent groups of data, especially when there is a reason to believe that both the groups have unequal variances or unequal sample sizes. In this part of the study, it was important to perform the t-test on the Pre and Post COVID period's monthly patient flow to check if there is any significant difference in both averages of the data.

Text Analysis using Clustering for the Presenting Complaints:

Building upon the quantitative insights gained from the exploratory data analysis, sought to delve into the qualitative aspects like patient issues or complaints and their categories to optimize the resource allocations, triage processes, and overall patient care in the emergency department.

As there were lot of complaints with various descriptions for each category in the "Presenting Complaint".

Step 1: Text Cleaning: The text was standardized by converting all the letters to lowercase, removing punctuation, numerical values, and common stop words (e.g.: "and"," the", "is") to focus the meaningful words.

Step 2: Tokenization: The cleaned text was split into individual words to support the analysis.

Step 3: Stemming and Lemmatization: All the words here were reduced to their root forms to ensure that the different variations of words are treated uniformly.

Step 4: Creating a Document: Term Matrix (DTM): A DTM was generated to quantify the frequency of all the words across the presenting complaints. This serves as a foundation to identify the patterns and similarities between different complaints.

Step 5: Clustering with K-means Algorithm: The K-means algorithm was applied to the DTM to group all the similar complaints together, each cluster representing different category prevalent among the patient presentations.

Step 6: Visualization: Word clouds were created for each cluster to visualize the most frequent terms and provide an intuitive understanding of the common themes within the groups.

Transition to Modelling: Understanding Patient Dynamics and Predictive Analysis:

After thoroughly analysing the dataset through extensive exploratory data analysis, Pre and Post COVID scenario changes and text analysis using clustering technique, I have developed a comprehensive understanding of the key variables and patterns influencing the patient flow and the operations in the emergency departments by examining and gaining insights of the patient arrival trends, breach patterns, presenting complaints and other complexities involved in managing the ED resources effectively.

This foundation of knowledge has set the stage for predictive modelling, in which I aimed to forecast the critical aspects of ED operations, such as daily patient arrivals, admission likelihood and waiting times. These models will be instrumental in anticipating demand, optimizing resource allocation, and ultimately enhancing patient care.

The modelling is divided into three key parts:

1. **Predicting Daily Patient Arrivals Using a Poisson Regression Model:** In this initial step I focused on forecasting the number of patients expected to arrive each day, enabling better staffing and resource management.
2. **Estimating Admission Likelihood:** Building on the arrival predictions, I decided to generate models that will estimate the probability that a patient will be admitted based on various factors, allowing for more efficient bed management.
3. **Predicting Waiting Times for Admitted Patients:** Finally, I generated this model which will predict the waiting times for patients likely to be admitted, helping to identify potential bottlenecks and improve patient throughput.

All these models serve a specific purpose within the broader context of ED management, contributing to a data-driven approach that enhances decision-making and operational efficiency. In the following sections we delve into the methodology, implementation and interpretation of these predictive models starting with

Models for predicting daily patient arrivals:

Model 1 - Poisson regression:

Poisson regression is a type of generalized linear model (GLM) used to model count data (Hausman et al. 1984). I assumed the response variable as the count of patient arrivals, follows a Poisson distribution. The model is designed to predict the number of occurrences of an event (patient arrivals) over a fixed period.

The general formula for Poisson Regression:

The model is represented as:

$$\log\big(E(y)\big) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Here in this case, it is:

$$\log\,(E(patient\ count) = \beta_0 + \beta_1\,(ED\ Arrival\ date)$$

Where:

$E(y)$ is the expected value of the **dependent variable** which here is our **patient count**

y is the count of the dependent variable.

β0 is the intercept.

β1 is the coefficient for the independent variable which is ED Arrival date (X1, X2…Xn).

Implementation:

Data Aggregation: The dataset was first aggregated to calculate the total number of patient arrivals for each day. This was done to get the precise count of the patients coming in daily till date for modelling purposes.

Train-Test Split: The dataset was split into a training set (80%) and a testing set (20%) to evaluate the model's performance. This approach ensures that the model is evaluated on unseen data, simulating real-world use cases.

Poisson Model Training: A Poisson regression model was fitted to the training data, with the count of patient arrivals as the dependent variable(y) and the ED arrival date as the independent variable(β1).

Model Testing and Future Predictions: The model was used to predict patient counts for both the test data and future dates. Predictions were made for specific future dates to forecast patient arrival volumes.

After analysing the poisson regression model results though I got some valuable insights and realised that it may not have handled the overdispersion adequately as the variance exceeds the mean. Usually, in the emergency department the fluctuations in the patient flow like number of arrivals on different days is very common which might be the reason behind the overdispersion. To address this issue I have applied the negative binomial model as an extension of the poisson model to check the predictions and handle the overdispersion issue effectively.

Model 2 – Negative Binomial Model:

The **Negative Binomial (NB) regression model** is an extension of the Poisson model that incorporates an extra parameter to handle the overdispersion. The model allows the variance to be greater than the mean, which is useful when modeling count data that exhibits overdispersion (variance > mean).

The general formula for a Negative Binomial regression is:

The model is represented as:

$$\log\big(E(y)\big) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Here in this case, it is:

$$\log\,(E(patient\ count) = \beta_0 + \beta_1\,(ED\ Arrival\ date)$$

Where:

$E(y)$ is the expected value of the **dependent variable** which here is our **patient count**

y is the count of the dependent variable.

β0 is the intercept.

β1 is the coefficient for the independent variable which is ED Arrival date (X1, X2…Xn).

However, unlike the poisson model here the negative binomial model introduces an extra theta term to the simple model to handle the overdispersion issue by adjusting the variance as:

$$Variance(y) = E(y) + \frac{E(y)^2}{\theta}$$

Implementation:

Step 1: I have used MASS package from which "glm.nb" function is used to fit a negative binomial regression model where ED Arrival Date as a independent variable and Patient Count as a dependent variable, also including an extra parameter to control for variance beyond the mean

Step 2 : The trained model was then used to predict the count of patients on the test data to evaluate the performance of the model on the unseen data.

Step 3: A dataframe for predictions of the future dates was created and the patient count were predicted for those dates.

Then I have moved on to the steps to generate more models and get best fit model for predictions and decided to generate a prophet model on my data.

Model – 3 Facebook Prophet Model:

The **Prophet Model** is an additive time-series forecasting model developed by Facebook. It is designed to handle time series data that may have seasonal patterns, trends, and other effects such as holidays or special events. Prophet is particularly effective in settings where there are regular cycles (e.g., daily, weekly, yearly), and it can manage missing data and outliers. It's also robust to abrupt changes in time-series data, making it ideal for forecasting in uncertain environments like healthcare(Taylor and Letham 2018).

Formula of the Prophet Model:

Prophet uses a generalized additive model to predict future data points. The basic formula for the Prophet model is:

$$\boldsymbol{y(t) = g(t) + s(t) + h(t) + \epsilon_t}$$

Where:

$\boldsymbol{y(t)}$ **:** The predicted value (in this case, daily patient count) at time t.

$\boldsymbol{g(t)}$**:** The trend component, representing long-term changes in the data (e.g., increasing or decreasing patient counts over time).

$\boldsymbol{s(t)}$ **:** The seasonality component, representing recurring patterns in the data (e.g., weekly or yearly cycles in patient arrivals).

$\boldsymbol{h(t)}$ **:** The effect of holidays or other events that might cause deviations from the usual pattern.

$\boldsymbol{\epsilon_t}$ **:** The error term representing any unexplained variability or noise in the data.

Components Explained:

Trend Component ($\boldsymbol{g(t)}$): Captures non-periodic changes in the data, such as long-term growth or decline. Prophet allows for two types of trends:

Linear trend: A simple linear increase or decrease.

Logistic growth: Growth that saturates at some maximum capacity.

Seasonality Component ($\boldsymbol{s(t)}$): Represents repeating patterns that occur at fixed intervals (e.g., daily, weekly, yearly). The seasonality component is modeled as a Fourier series, allowing Prophet to capture complex repeating patterns.

Holiday Effects ($h(t)$): The model can include the effects of holidays or special events, which may cause spikes or drops in the data. These are user-specified and added to the model to account for deviations from the regular trend and seasonality patterns.

Error Term (ϵ_t): This represents the residuals or noise in the data that the model cannot explain.

Implementation of the Prophet model:

Data Preparation:

I first preprocessed the data by renaming the key variables as :

ds : This represents the ED Arrival date in the dataset which will used for the predicitons of future dates.

y : This represents the patient_count per day arriving in the emergenct department which is the dependent variable which needs to be predicted using the model considering various factors.

Model fitting:

The Prophet model was fitted on the prepared dataset, with the date as the independent variable (ds) and patient arrivals as the dependent variable (y). Prophet works by decomposing the time series into three components:

Trend: Long-term increase or decrease in the patient flow.

Seasonality: Periodic patterns such as daily, weekly, or yearly variations in patient flow.

Holidays or Events: Impact of specific events like holidays or public events on patient flow.

Making future predictions:

Using the fitted model, I made the future predictions for the next 10 days to check the average count of patients coming with this model that included **yhat** (predicted patients count) , **yhat_lower and yhat_**upper which are the lower and upper bound predictions which also represents the uncertainty of the forecasted outputs.

Visualising the predictions of the Model:

After getting the predictions of the patient count for next 10days I have decided to plot a line graph to visualize the predicted values using the ggplot function.

Models for Estimating Admission Likelihood:

In this section of the analysis, we aim to predict the likelihood of patient admissions in the Emergency Department (ED) daily. Building upon the patient arrival predictions generated using the Poisson, Negative Binomial, and Prophet models, I've chosen to use the Poisson model predictions as the basis for this part of the study. Although all three models provide similar patient count predictions, I have opted for the Poisson model's prediction due to its simplicity and interpretability in count data modelling. Additionally, two key variables, admission status and ED Area, were incorporated to generate a model that would forecast the admission likelihood of patients entering the ED each day. In this section, we will develop a logistic regression model to predict whether a patient will be admitted based on these factors.

Model – 4 Logistic Regression Model:

Logistic regression is a statistical model used to predict a binary outcome based on one or more predictor variables. In this case, the binary outcome is whether a patient is admitted or not. Logistic regression estimates the probability that a particular event occurs (in this case, a patient being admitted) and fits a curve that represents this probability between 0 and 1(Berkson 1944).

The general form of a logistic regression model is:

$$\log\left(\frac{P(y-1)}{1-P(y-1)}\right) = \boldsymbol{\beta_0 + \beta_1 X_1 + \beta_2 X_2} + \cdots + \boldsymbol{\beta_n X_n}$$

Where:

$\boldsymbol{P(y-1)}$ is the probability that the patient is admitted (admission = 1).

β0 is the intercept.

β1, β2,…,βk are the coefficients for the predictor variables **X1,X2,...,Xk** .

In this model, the dependent variable is whether a patient is admission status, and the independent variables include the predicted Poisson count (i.e., the estimated number of patients arriving) and ED area type (categorized into different levels based on patient care priority, such as "Majors", "Minors", etc.).

Implementation of the Logistic regression model:

The logistic regression model was then fitted using the "glm()" function, specifying the "binomial family" for logistic regression. This model predicts the likelihood of a patient being admitted based on their "ED Area" Type and predicted Poisson count.

Data preparation: Converting all the variables into numerical format and then categorizing the **"ED Area"** into four types as **Type – 1 "Majors", Type – 2 "Minors", Type – 3 "Resus", Type – 4 "Paediatrics"** which will be our independent variable. This helps us to predict the likelihood of admission of different types of patients in the ED.

Converting the **Admission status** column into a binary variable by changing the admitted to 1 and not admitted 0 as it our **dependent variable** in the model.

Including the Poisson model predictions: I have included the predictions given by the Poisson model from the earlier section where I predicted the daily patient arrival count as one of the variables for this model from which we will get the predictions again for the number of people who are more likely to be admitted in the hospital on daily basis.

Multicollinearity check: I performed variance inflation factor analysis before fitting the logistic model to ensure that there is no multicollinearity issue among the variables used before fitting the model.

Predicting the Likelihood for 3 months: Prediction of admission likelihood for the next three months was generated using the logistic model which is from July 2024 to September 2024 using the Poisson prediction patient count, Ed-area and the admission status.

The predictions were made according to the type of the patient, which is type 1,2, and 3 as they are top 3 ones among the four categories.

Visualising the Model characteristics and predictions in graphs or plots: A bar plot was created to visualize the average predicted admission likelihood for each month (July - September 2024) and ED area type. This plot helps illustrate the differences in admission probabilities across the different ED areas and time periods.

Two diagnostic plots were generated to check the model performance on the data to evaluate the goodness of fit. One was a calibration plot which shows the curve that tells us how well the model has given the predictions match probabilities of admissions. Second one was a density plot that predicted the admission likelihood distribution among the admitted and non-admitted patients.

Generating the ROC curve and AUC calculation: To check the model performance with various thresholds, I tried to visualise it graphically and understand the sensitivity and specificity trade-off in this admission probabilities scenario.

Then to check the overall accuracy and ability, I calculated the AUC values when close to 1 indicate excellent model performance and while values closer to 0.5 suggest a model with no discriminatory power.

Modelling for predicting the Waiting Time in the Emergency Department:

Given the critical importance of managing waiting times in emergency departments (ED), predicting the time patients are waiting for someone to see and start the treatment for their issues is a primary focus of this analysis. In the previous sections, exploratory analysis revealed significant insights into patient arrivals and patterns in the ED, such as patient surges on Mondays and comparitively lower volumes on weekends or other days of the week. This led to the hypothesis that waiting times could also vary considerably depending on different days of the week. Thus, in the next phase of the analysis I focused on predicting waiting times particularly the **"Arrival to Seen" minutes** on **"Mondays", "Weekdays", and "Weekends",** by considering **"ED Area"** as a variable segregating the patient types.I chose to start with time series analysis, looking at the patterns in the data to delve depper by knowing the properties like seasonality, stationarity, outliers etc., present in the dataset.

Implementating the Time series on the Data:

Data preparation: Transforming the data into the timeseries format by using the ts() function by using the forecast package. As this will help to do the time series analysis by ensuring that all the observations are are treated in sequential order with a frequency set to 365 to reflect the daily data over the given years.

Decomposition of Timeseries: Decomposition plot helps us to get the viasualisation of the essential components like trend, seasonality and residual for a clearer understanding of the patterns.

ACF and PACF plots:

Autocorrelation Function (ACF): It measures the correlation between a time series and lagged versions of itself at different time steps. It quantifies the degree of similarity between observations at a given time and observations from earlier time periods (lags). Essentially, the ACF tells us how the values in the time series are related to their previous values, helping identify repeating patterns or seasonality.

Formula:

The autocorrelation at lag k is computed as:

$$\boldsymbol{\rho_k} = \frac{\sum_{t=k+1}^{n} (yt-\hat{y})(yt-k-\hat{y})}{\sum_{t=1}^{n} (yt-\hat{y})^2}$$

Pk is the autocorrelation at lag k,

yt is the value of time series at time t,

$\hat{y}$ is the mean of the time series,

n is the number of observations in the series.

Partial Autocorrelation Function (PACF): It measures the correlation between a time series and lagged versions of itself, after removing the effects of all shorter lags. While the ACF considers the overall correlation with all previous lags, the PACF isolates the direct effect of a given lag, controlling for the influence of earlier lags.

Formula:

The partial autocorrelation at lag kkk is the residual correlation after accounting for the effects of intermediate lags. It can be expressed using regression:

$$\phi_k = \text{Corr}(,yt-k|yt-1,yt-2,...,yt-k+1)$$

Where:

ϕ_k is the partial autocorrelation at lag k,

y_t is the value at time t,

y_{t-k} is the value at lag k.

Hence, I plotted the ACF and PACF plots to check the autocorrelations and partial autocorrelations between the current values and the lagged values present in my timeseries data that reveals the patterns or potetial seasonalities. Meanwhile, the PACF plot was used to determine the extent of correlation between the current values and specific lagged values after accounting for the influence of earlier lags, helping to isolate significant lag orders that may be critical in the forecasting model. These plots provide important insights into the underlying time series structure and guide the selection of appropriate lag terms for modeling.

3.Sationarity check: I performed a statistical test called Augmented Dickey Fuller (ADF) test to check if there is any stationarity or contains a unit root (no stationarity) in the data. Stationarity basically means the timeseries data having constant statistical properties like mean, variance and autocorrelation over time otherwise it is considered non stationary.

Null Hypothesis (H_o): The time series has a unit root, meaning it is non-stationary.

Alternative Hypothesis (H_1): The time series does not have a unit root, meaning it is stationary.

P-Value: If the p-value is below a chosen significance level (commonly 0.05), we reject the null hypothesis, indicating that the time series is stationary.

Test Statistic: Compare the test statistic with the critical values at different confidence levels (1%, 5%, and 10%). If the test statistic is less than the critical value, we reject the null hypothesis, suggesting the series is stationary.

It is very crucial step to determine the stationarity of the data before you apply the models for forecasting the future so that we could achieve the stationarity using some techniques if it is not stationary. After all the test and the properties of the data were checked I moved on to implement the Seasonal Arima model as it had season,trend and stationarity in the timeseries data.

Model – 5 Seasonal ARIMA Model:

Based on these observations, I decided to apply a **Seasonal AutoRegressive Integrated Moving Average (SARIMA)** model. This model was selected because it effectively handles seasonality and trends, as well as the time-dependent fluctuations seen in patient flow. Time series analysis enables us to account for the inherent seasonality in the data, as well as the weekly and long-term trends that may affect patient waiting times across different periods(Box George et al. 1976).

SARIMA Model Definition:

SARIMA is an extension of the ARIMA model that incorporates seasonality into time series forecasting. The general SARIMA model is written as:

$$\boldsymbol{SARIMA\ (p, d, q)(P, D, Q)_m}$$

Where:

$(p, d, q)$are the non-seasonal parameters.

$(P, D, Q)$are the seasonal parameters.

m is the periodicity of the seasonal pattern.

The model assumes that the underlying time series is affected by past values (autoregression), previous forecast errors (moving average), and seasonal patterns.

Implementation of the SARIMA MODEL:

Data Preparation: The dataset is first prepared by creating a new column, Day_Type, which categorizes each record based on the day of the week into "Monday," "Weekend" (Saturday and Sunday), and "Weekday" (all other days).

Additionally, the Admitted_Flag variable is converted into a factor to represent whether a patient was admitted (1) or not (0).

Fitting the SARIMA Model: The time series data (sarima_ts) is then passed through the auto.arima() function to automatically fit a Seasonal AutoRegressive Integrated Moving Average (SARIMA) model.

This function identifies the best parameters for the model based on the data's seasonality, trend, and autoregressive patterns.

Forecasting the Next 14 Days: The SARIMA model is used to forecast the **"Arrival to Seen"** minutes for the next 14 days (2 weeks ahead). This prediction helps in identifying trends in patient waiting times.

Visualizing Historical Data and Forecast: A combined data frame is created to plot both the historical time series data and the forecasted values, along with confidence intervals.

Using the ggplot2 package, a time-series plot is generated that shows the historical data, the forecasted values, and a shaded area representing the 95% confidence intervals for the forecast.

After implementing the SARIMA model and analyzing the waiting times in the emergency department (ED), there were noticeable limitations in its predictive capacity, particularly with handling non-linear relationships, sudden variations, and specific feature interactions. Despite capturing some trends and seasonal patterns, SARIMA struggled to effectively account for the complex and dynamic nature of ED operations, such as varying day types (Mondays, Weekends, Weekdays) and the time-of-day effect on waiting times. Therefore, I decided to explore another modeling approach that could potentially capture these complexities better.

Model – 6 Random Forest Model:

Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve the overall predictive accuracy and reduce overfitting of the model on the data. Each decision tree is built on a random subset of features and data points, and the final prediction is obtained by averaging the outputs of all trees for regression or taking the majority vote for classification(Breiman 2004).

The general formula for Random Forest regression is:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} f_t\,(x)$$

Where:

T is the total number of decision trees in the forest.

$f_t\,(x)$ is the prediction from the ttt-th decision tree.

x is the input feature vector.

Random Forests are particularly useful when there are complex and many features, which is the case in predicting ED waiting times, where factors like “Day Type”, “Hour of the day”, and “Admission status” all play a very important role.

Implementation of the Model:

Data Preparation: To build the Random Forest model, I first prepared the dataset by creating a new feature, **“Day_Type”,** which categorizes each day into **"Monday", "Weekend" (Saturday and Sunday), or "Weekday" (all other days).** The dataset also includes features like **“Hour of the Day”** to capture the time patterns and **“Admitted_Flag”** which is if the patient was admitted or not.

The target variable for this model is **“Waiting_Time”,** which represents the number of minutes between a patient’s arrival and the time they were first seen by a doctor.

Random Forest Model Training: The Random Forest model was trained using 100 trees. The features used to predict **Waiting_Time** include **“Day_Type”, “Hour of the Day”, and “Admitted_Flag”.** The model works by averaging predictions across all trees, allowing it to capture complex interactions between these features.

Predictions and Results: Predictions were made on the test data to estimate the waiting times based on the trained Random Forest model. The model's predictions were grouped by **“Day_Type” and “Hour of the Day”** to calculate the average predicted waiting times for different time periods and day types.

Visualising the results: The average predicted waiting times were plotted to visualize how waiting times vary across different “**hours of the day”** for “**Mondays”, “Weekdays”, and “Weekends”**. This helps to identify patterns in patient waiting times that could assist in better resource allocation in the emergency department.

Chapter 5

Results:

The results we got for the Pre-Covid and Post-Covid were as follows:

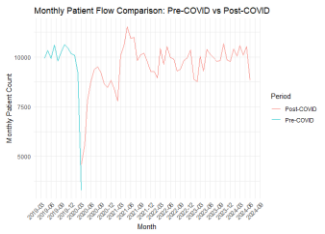

Fig.25 Monthly pattern

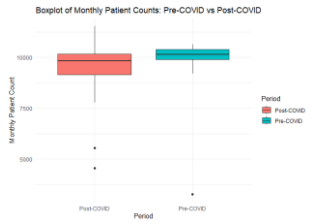

Fig.26 Box Plot Pre-Post

```
        Welch Two Sample t-test

data:  monthly_arrivals_pre$Count and monthly_arrivals_post$Count
t = 0.037399, df = 12.853, p-value = 0.9707
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1286.396  1331.665
sample estimates:
mean of x mean of y
 9556.500  9533.865
```

Fig.27 T-test Results

Monthly patient flow comparison in the **fig 25** demonstrates a sharp decline in the patient count in 2020 which was during the pandemic would probably be due to the obvious the reasons like fear of virus transmission and lockdown all over world. However, after that period the data shows a gradual recovery in the patient flow by stabilizing the patterns to the same level of pre-covid which indicates that the situation turned normal as the pandemic progressed over the time.

Then the boxplots in **fig26** were plotted to visualise the pattern in both the periods which shows that the distributions are similar in both, with overlapping medians and interquartile ranges. Although there are some outliers in the post-covid period which indicates some fluctuations occasionally in the patient flow.

The results of Welch's t -test were as follows – A Welch Two-Sample t-test compare the Pre and Post COVID periods in fig 27 yielded a **p-value = 0.9707**.

Null hypothesis (H0) – There is no significant difference in mean of the patient arrivals in both the periods (i.e. $\mu\, pre = \mu\, post$).

Alternative hypothesis(H1) – There is a significant difference in the mean of the patient arrivals in both the periods (i.e. $\mu\, pre \neq \mu\, post$).

This p-value = 0.9707 > 0.05 indicates that there is **no significant difference statistically** between the two periods. It suggests that over time the overall patient volume in the ED has normalized, and there has been no lasting impact on the average monthly patient flow due to the pandemic.

While the COVID-19 pandemic initially led to substantial disruptions in patient arrivals at the ED, likely resulting in the postponement of numerous planned operations and surgeries, the department's subsequent recovery demonstrates the hospital's flexibility and effective adaptability. This resilience

in healthcare management during and after the global crisis highlights the institution's capacity to navigate and respond to unprecedented challenges.

The Results Text Analysis using clustering are as follows:

The clustering process revealed 3 primary groups of presenting complaints with different categories.

Fig.28 Cluster 1

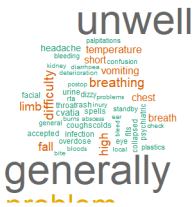

Fig.29 Cluster 2

Fig.30 Cluster 3

Cluster 1: Pain Related Complaints

Dominant Terms: "pain," "chest," "abdominal," "head," "back".

This cluster consists of patients with various types of pains like chest, abdominal pains and headaches. These complaints often require prompt assessment to rule out critical conditions like myocardial infarctions or internal injuries.

Cluster 2: Generally Unwell

Dominant terms: "unwell," "general," "feeling," "dizzy," "tired"

Patients in this cluster report non-specific symptoms indicating a general feeling of being unwell. These cases may necessitate comprehensive diagnostic evaluations to identify underlying causes, which can contribute to longer ED stays.

Cluster 3: Injury and Trauma Related complaints

Dominant Terms: "injury," "fall," "wound," "fracture," "bruise"

This group includes patients presenting with injuries resulting from falls, accidents, or other traumatic events. Efficient management of these cases is essential to prevent complications and ensure timely treatment.

Implemented Models:

Results for Model 1 - Poisson Model:

```
Call:
glm(formula = patient_count ~ ED.Arrival.Date, family = poisson,
    data = train_data)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)     5.352e+00  6.104e-02  87.677  < 2e-16 ***
ED.Arrival.Date 2.142e-05  3.253e-06   6.582 4.63e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 12846  on 1531  degrees of freedom
Residual deviance: 12803  on 1530  degrees of freedom
AIC: 24416

Number of Fisher Scoring iterations: 4
```

Fig.31 Poisson Summary

ED.Arrival.Date <date>	predicted_patient_count <dbl>
2024-08-02	323
2024-08-03	323
2024-08-04	323
2024-08-05	323
2024-08-06	323
2024-08-07	323
2024-08-08	323
2024-08-09	323
2024-08-10	323

...

Fig.32 Poisson Predictions

Intercept: The intercept = 5.352 represents the log count of patient arrivals on the earliest date in the dataset. Since Poisson regression models log counts, we interpret this as follows: Expected count of patient arrivals = exp (5.352) ≈ 211.61. This means that, the expected number of patient arrivals is about **211 patients per day**.

ED Arrival Date Coefficient: This coefficient **2.142e-05** is very small but positive, indicating that the count of patients slightly increases over time. This expected rise in patient arrivals each day is: Expected increase = exp(2.142e−05) −1 ≈ 0.00002142%. This suggests a minor but steady increase in patient numbers over time. Specifically, log count of patient arrivals increases by **0.00002142%**.

Statistical Significance (p-values): Both the intercept and the ED arrival date coefficient have extremely small p-values which is **less than 2e-16** and **4.63e-11** respectively, indicating that both are highly statistically significant. This means that the values are not equal to zero, and the variables of this model are explaining the variation in patient arrivals significantly.

Null Deviance (12846) vs. Residual Deviance (12803): Null deviance represents how well the model with only the intercept with no other predictors fits the data. **Residual deviance** represents how well the model with the predictors fits the data. The slight reduction in deviance from 12846 to 12803 indicates that adding the ED arrival date to the model as a predictor slightly improves the model's fit, though the performance improved is very moderate.

Degrees of Freedom (1531, 1530): The null degrees of freedom represent the total information available for us to estimate the parameters of the model, which includes only intercept and here the value is (1532 -1) = 1531.

The residual degree of freedom shows us the room left for the model to explain the variation in our "patient count" (dependent variable) considering the ED Arrival date as predictor along with the intercept. Here the value of the residual degree of freedom is (1532 – 2) = 1530.

AIC (24416): Akaike Information Criterion (AIC) is a measure of the model's goodness-of-fit, penalized for complexity. Lower AIC values indicate a better fit, with a balance between model accuracy and simplicity. In this case, an AIC of 24416 gives a benchmark for comparing this model to more complex models.

The Residual deviance is 12803 and the degrees of freedom is 1530. As we know that the values of residual deviance must be close to degrees of freedom if the model is a good fit for the data. However, the deviance is indeed much larger than the degrees of freedom value, which is the evidence of overdispersion issue.

Patient Arrival Predictions: The predicted patient counts for specific future dates (August 2024) show stable values around 323 patients per day. This suggests that the ED can expect consistent patient volumes for the upcoming week.

Sample predictions: August 2, 2024: 323 patients, August 3, 2024: 323 patients, August 4, 2024: 323 patients

These predictions can be used by hospital management to allocate resources efficiently, ensuring that staffing levels and medical supplies are aligned with expected patient volumes.

Model for Extended Prediction: The model was further extended to predict patient arrivals for the next three months as the longer-term forecasts will provide us the insights into future patient flow, which helps in more strategic planning. The results for next three months were also quite similar.

Results for Model -2 Negative Binomial Model:

```
Call:
glm.nb(formula = patient_count ~ ED.Arrival.Date, data = train_data,
    init.theta = 41.27282393, link = log)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)     5.357e+00  1.794e-01  29.864   <2e-16 ***
ED.Arrival.Date 2.114e-05  9.563e-06   2.211    0.027 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(41.2728) family taken to be 1)

    Null deviance: 1551.5  on 1531  degrees of freedom
Residual deviance: 1546.6  on 1530  degrees of freedom
AIC: 16455

Number of Fisher Scoring iterations: 1


              Theta:  41.27
          Std. Err.:  1.69

 2 x log-likelihood:  -16448.53
```

Fig.33 Negative Binomial Summary

ED.Arrival.Date <date>	predicted_patient_count_nb <dbl>
2024-08-02	323
2024-08-03	323
2024-08-04	323
2024-08-05	323
2024-08-06	323
2024-08-07	323
2024-08-08	323
2024-08-09	323
2024-08-10	323

Fig.34 NB Predictions

Intercept: The intercept = 5.357 represents the log count of patient arrivals on the earliest date in the dataset. Since, negative binomial model log counts, we interpret this as follows: Expected count of patient arrivals = exp (5.357) ≈ 212.05. This means that, the expected number of patient arrivals is about **212 patients per day**. This means the expected count of patients for the earliest date in the dataset is approximately 212 patients per day.

ED Arrival Date Coefficient: This coefficient **2.114e-05** is very small but positive, indicating that the count of patients slightly increases over time. This expected rise in patient arrivals each day is: Expected increase = exp(2.114e−05) −1 ≈ 0.00002114%. This suggests a minor but steady increase in patient numbers over time. Specifically, log count of patient arrivals increases by **0.00002114%**.

Statistical significance (p-values): Both the intercept and the ED arrival date coefficient have statistically significant p-values. The intercept has an extremely small p-value of less than 2e-16, indicating a very strong significance. The ED arrival date coefficient has a p-value of 0.027, which is also statistically significant as it less than the $\alpha = 0.05$ and proves that the "ED Arrival date" (independent variable) has a significant effect on the "patient_count" (dependent variable).

Null Deviance (1551.5) vs. Residual Deviance (1546.7): Null deviance represents how well the model with only the intercept with no other predictors fits the data. **Residual deviance** represents how well the model with the predictors fits the data. The moderate reduction in deviance from 1551.5 to 1546.6 indicates that adding the ED arrival date to the model as a predictor slightly improves the model's fit, though the performance improved is very moderate.

Degrees of Freedom (1531, 1530): Like the Poisson model**,** the null degrees of freedom represent the total information available for us to estimate the parameters of the model, which includes only intercept and here the value is (1532 -1) = 1531. The residual degree of freedom shows us the room left for the model to explain the variation in our "patient count" (dependent variable) considering the ED Arrival date as predictor along with the intercept. Here the value of the residual degree of freedom is (1532 – 2) = 1530.

AIC (16455): As we know that the lower AIC value indicates better fit, In this case when we compare this model with the Poisson model we could see that the negative binomial model provides significantly better fit, specially by handling the overdispersion slightly better. This model's reduced AIC value provides evidence that the balance between accuracy and simplicity is better here.

Theta (ϴ): The estimated dispersion parameter (41.27) suggests that overdispersion is indeed present, which validates my idea of using the negative binomial model over the simple poisson model though it is not that extreme.

Sample predictions: August 2, 2024: 323 patients, August 3, 2024: 323 patients, August 4, 2024: 323 patients

These predictions can be used by hospital management to allocate resources efficiently, ensuring that staffing levels and medical supplies are aligned with expected patient volumes.

Model for Extended Prediction: The model was further extended to predict patient arrivals for the next three months as the longer-term forecasts will provide us the insights into future patient flow, which helps in more strategic planning. The results for next three months were also quite similar.

Though the Negative binomial model gave us the results handling the overdispersion better than the Poisson model, the predictions from both the models were same which is 212 patients per day. Hence, I have decided to apply more models to get better predictions.

Results for Model-3 Prophet Model:

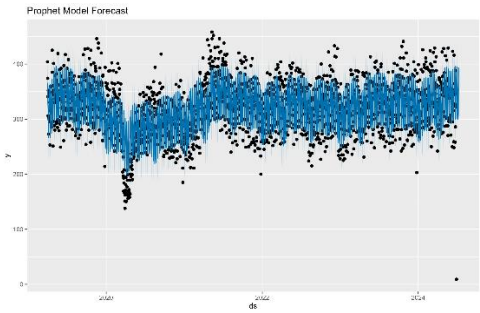

Fig.35 Prophet Predictions

ds <S3: POSIXct>	yhat <dbl>	yhat_lower <dbl>	yhat_upper <dbl>
2024-07-03	359	321	400
2024-07-04	353	318	393
2024-07-05	347	308	387
2024-07-06	299	258	335
2024-07-07	308	268	347
2024-07-08	394	351	432

Fig.36 Prophet Predictions

Predicted values for the next 10 days using the prophet model :

Sample forecasted value:

August 2, 2024: Expected Patient Count = 315, (Lower Bound = 305, Upper Bound = 325)

August 3, 2024: Expected Patient Count = 316, (Lower Bound = 306, Upper Bound = 326)

August 4, 2024: Expected Patient Count = 318, (Lower Bound = 308, Upper Bound = 328)

August 5, 2024: Expected Patient Count = 320, (Lower Bound = 310, Upper Bound = 330)

From the forecasts given by the Prophet model, we can see that patient counts are consistent including the lower and upper bound values which indicates that it is very beneficial in making the decisions for staff and resources allocation including other decision making for the management.

The graph produced by the the prophet model shows us the daily fluctuations in the patient count daily and also captures the seasonal pattern accurately for the future.

Model Performance or goodness check: Having said that the prophet model is used here as it captures the seasonal and trend components well and gives us the predictions, we can see that the flow of patients is steady along with displaying the uncertainity bounds which suggests that the model gives us a confident forecast about the possible deviations due to any reasons on daily basis in the patient count.

Likelihood prediction:

Results for Model – 4 Logistic regression Model

```
Call:
glm(formula = Admitted_Flag ~ ED.Area.Type + predicted_poisson_count,
    family = binomial, data = Likelihood_model_data)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)             -1.214e+01  2.485e-01  -48.85   <2e-16 ***
ED.Area.TypeType-2      -2.225e+00  7.706e-03 -288.73   <2e-16 ***
ED.Area.TypeType-3       5.404e-01  2.304e-02   23.45   <2e-16 ***
ED.Area.TypeType-4      -2.379e+00  9.604e-03 -247.66   <2e-16 ***
predicted_poisson_count  4.044e-02  7.848e-04   51.53   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 842904  on 610438  degrees of freedom
Residual deviance: 682693  on 610434  degrees of freedom
AIC: 682703

Number of Fisher Scoring iterations: 4

             (Intercept)       ED.Area.TypeType-2       ED.Area.TypeType-3
            5.344667e-06            1.080546e-01            1.716621e+00
      ED.Area.TypeType-4 predicted_poisson_count
            9.268052e-02            1.041272e+00
```

Fig.37 Logistic Summary

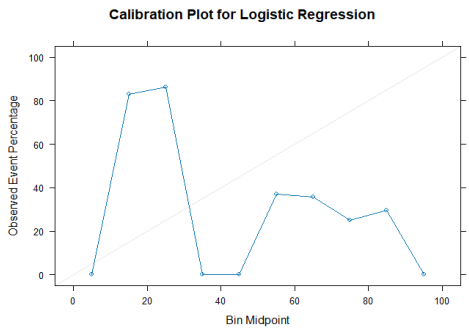

Fig.38 Calibiration Plot

Coefficients: The coefficients for **ED.Area.Type** (Type-1, Type-2, Type-4) are all negative, indicating that patients in these areas have lower odds of being admitted compared to the Type-3, which represents the "Resus" area. The intercept for Type-3 “Resus” is 5.404 which shows that it is extremely likely to be admitted to the hospital. “Majors” are the next category of patients who are likely to be admitted after the “Resus”.

Predicted poisson count has a very small but positive coefficient (4.044e-04), indicating that as the predicted patient count increases, the chances of patient count being admitted slightly increase which is very obvious.

Statistical Significance: For all the coefficients the p-values are less than 2e-16, indicating strong evidence that these variables have a significant relationship with the admission likelihood.

Null Deviance (842904) vs **Residual Deviance** (682693): As we know that the Null Deviance is the value that gives the performance of the model with only intercept, we can see that it wouldn’t be the best one for prediciting the admission likelihood of the patients. On the other hand, the residual deviance which is the value of the model that is considering both intercept and predictors seems to be a better fit as it less than the null value.

AIC (682703): The Akaike Information Criterion (AIC) is 682703. As we know that the lower AIC value, the better the model fits the data, while considering model complexity. This value can be used for comparison with other models if needed.

Calibration Plot: The **Calibration Plot** shows how well the predicted probabilities align with actual observed admission events. The observed percentages fluctuate, especially in the lower bins, indicating some deviations from perfect calibration. However, this can still be useful for understanding how well the model aligns with reality in terms of event likelihood.

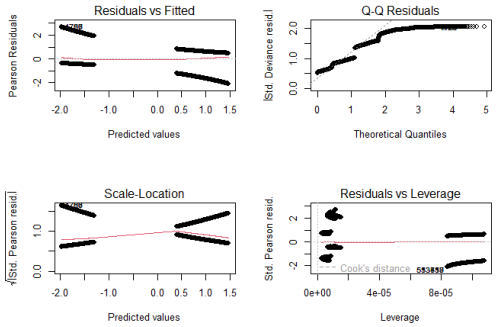

Fig.39 Diagnostic Plots

Diagnostic Plots:

Residuals vs Fitted: Shows no significant pattern, which suggests that the model is well-fitted to the data, with no major deviations.

Q-Q Residuals Plot: Slight deviations at the tail, which could suggest the presence of some outliers or skewness, but overall follows the expected pattern.

Scale-Location: The points are relatively horizontal, showing homoscedasticity, which indicates that the variance of residuals is constant.

Residuals vs Leverage: No significant points stand out with high leverage. Cook's distance suggests no overly influential points.

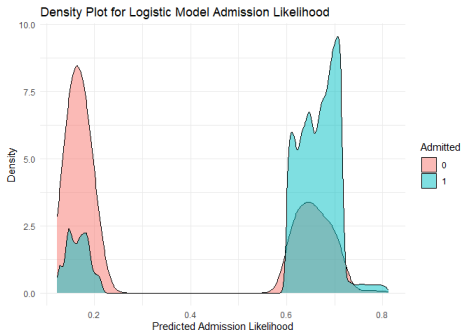

Fig.40 Density Plot

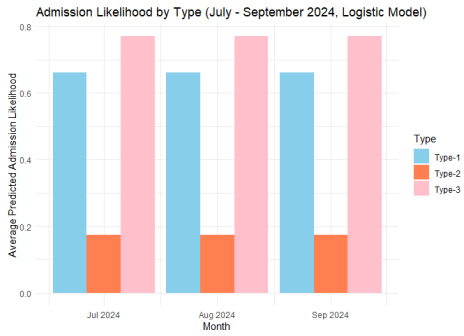

Fig.41 Admit-Likelihood for 3months

Density Plot:

The **Density Plot** reveals that the logistic model effectively separates admitted and non-admitted patients based on the predicted admission likelihood. The red area represents the patients who are not admitted and blue are represents the patients who are admitted with both the axis showing the density and prediction likelihood of the patients in the ED. Here, the patients who were not admitted has 0.2 or less, not admitted patients closer to 0.6 to higher and the overlap in between which is foe few patients aound 0.2 to 0.3 predicted likelihood.

Bar Plot - Admission Likelihood by Type:

Type-1 (Majors) in blue consistently shows higher predicted admission likelihood across all months, as expected since these patients are more likely to require admission based on their severity.

Type-2 (Minors) in orange shows the lowest likelihood, as these cases are generally less severe.

Type-3 (Resus) in pink with critical nature of these patients, shows the highest likelihood among all three, reflecting the urgency of care and treatment. This trend highlights how patient characteristics from different areas of the ED contribute to variation in the admission likelihoods.

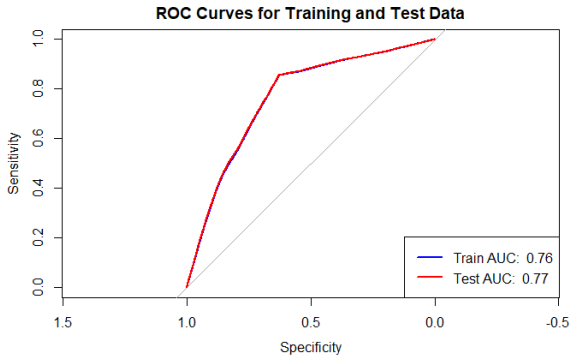

Fig.42 Logistic ROC Curve

ROC and AUC values : The diagonal line dividing the train and test data shows how well the model is performing in classifying the data into subsets based on the likelihood of admission.

AUC Values:

Training AUC: The Area Under the Curve for the training data is 0.76, indicating that the model has a good ability to distinguish between admitted and non-admitted patients on the training set.

Test AUC: The AUC for the test data is 0.77, showing that the model generalizes well to unseen data, with similar performance to the training set.

This shows that the model performs quite well on the unseen data and hence we can consider that the 77% of the randomly picked admitted patients are more likely to be admitted copared to the not admitted patients.

Waiting time predictions:

Results for Time-Series analysis:

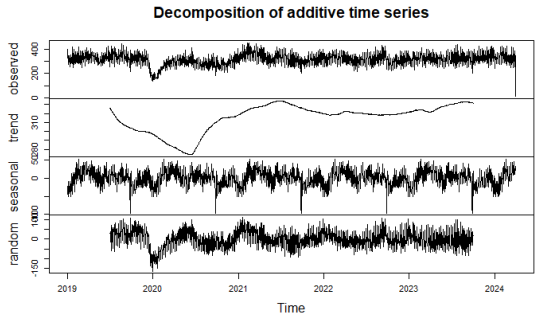

Fig.43 Decomposition plot

Trend: Shows a clear decrease during the COVID period and a subsequent gradual recovery post-pandemic. This indicates how patient arrivals dropped significantly during the pandemic but recovered over time.

Seasonal: Highlights the repetitive patterns in patient arrivals, which could be linked to weekly or monthly cycles in healthcare demand.

Residual: Captures the random fluctuations that cannot be attributed to trend or seasonality. Some peaks in this component could indicate extraordinary events impacting ED arrivals.

This Decomposition helped us in separating the essential elements of the time series, allowing us to model trend, seasonality, and noise separately for better forecasting.

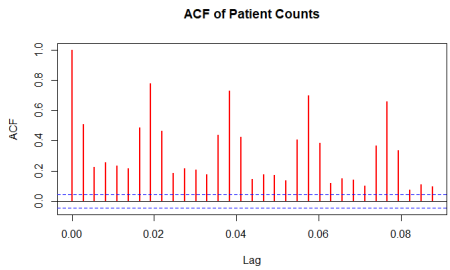

Fig.45 ACF Plot

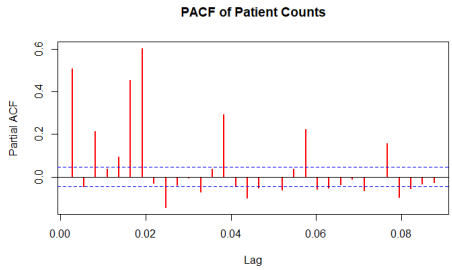

Fig.46 PACF Plot

ACF plot observations:

The ACF plot shows a strong autocorrelation at lag 1, which indicates that each patient count on a given day is highly correlated with the count from the previous day. The peaks around lags 0.02, 0.04, and 0.06 show significant autocorrelations at other regular intervals, suggesting **seasonality or periodicity** in the patient count data.

The correlations gradually decay, but there are still notable spikes at multiple lags, suggesting some long-term dependency in the series. The dashed blue lines represent the **confidence interval** (usually

at 95%). Any spike outside these lines represents statistically significant autocorrelation, while spikes within the lines are not significant.

PACF plot observations:

The lag 1 shows a significant spike, which means the current patient count is directly influenced by the count of the previous day. The PACF shows a few significant partial autocorrelations at earlier lags (lags around 0.02 and 0.03), but they decay more rapidly compared to the ACF plot.

The PACF suggests that the short-term dependencies are strong for the first few lags, and after controlling for these, later lags do not contribute much individually. his pattern often suggests that an AR(1) process (autoregressive model of order 1) might be a good starting point for the model, as the first lag has the strongest influence.

In conclusion, ACF suggests a strong correlation between patient counts on consecutive days and shows possible periodicity or seasonality in the data and on the other hand, PACF shows that after accounting for the first few lags, the correlations become negligible, which indicates the presence of short-term dependencies.
These insights suggest a model that incorporates short-term autoregressive terms (AR terms) and may need to account for potential seasonality. The significant lags in the ACF and PACF plots guide the selection of appropriate AR or MA (moving average) terms in time series models like ARIMA or SARIMA.

Then we have also performed the ADF test to check the statonarity which was as follows:

```
Warning in adf.test(daily_counts_ts, alternative = "stationary") :
  p-value smaller than printed p-value

        Augmented Dickey-Fuller Test

data:  daily_counts_ts
Dickey-Fuller = -4.3546, Lag order = 12, p-value = 0.01
alternative hypothesis: stationary
```

Fig.47
ADF Test

Based on the Augmented Dickey-Fuller test results, with a p-value of 0.01, we reject the null hypothesis and accept the alternative hypothesis, indicating that the time series is stationary.

Considering all the results from the ACF, PACF, and decomposition plots, we now know the presence of seasonality, trend, moving average components, and the stationarity of the data. Hence, I have chosen to implement the SARIMA model.

Results for Model – 4 SARIMA model:

Interpretations for the SARIMA model outputs:

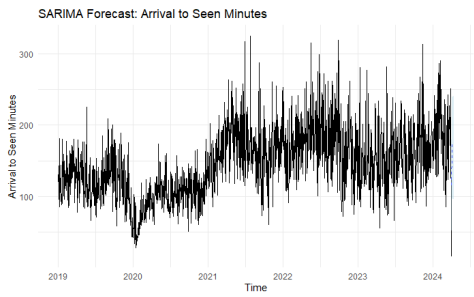

Fig.48 SARIMA Predictions

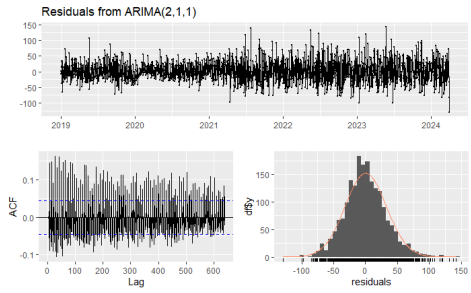

Fig.49 Residuals Plot

SARIMA Forecast (Arrival to Seen Minutes):

The SARIMA model forecast provides predictions for the "Arrival to Seen" waiting times in the emergency department over time. From the forecast plot, we can see a general increasing trend in waiting times, with periods of higher variability during specific time intervals.

There is noticeable fluctuation over the forecasted period, indicating that patient waiting times may vary significantly depending on certain days, likely influenced by patient volumes or operational factors in the hospital. The plot suggests that waiting times may exceed 200 minutes in the near future, particularly in 2023–2024, indicating that there could be capacity or resource allocation challenges in the ED during these periods.

Residual Analysis from ARIMA (2,1,1): This residuals plot shows no clear patterns or trends, indicating that the residuals are mostly random and that the ARIMA model has captured the underlying structure of the data well.

ACF plot (bottom-left): The autocorrelation plot of the residuals reveals that most of the autocorrelations fall within the confidence bands (dashed lines), suggesting that the residuals are mostly white noise. However, there are some slight correlations at certain lags, which may indicate minor remaining patterns in the data.

Distribution of Residuals (bottom-right): The histogram of residuals is approximately normally distributed, which confirms that the model residuals follow a roughly normal distribution, as expected for a well-fitted model. Overall, these diagnostics indicate that the ARIMA (2,1,1) model fits the data well, with minimal autocorrelation and residuals that resemble white noise.

```
        Ljung-Box test

data:  Residuals from ARIMA(2,1,1)
Q* = 2587.8, df = 380, p-value < 2.2e-16

Model df: 3.   Total lags used: 383
```

Fig.50 Ljung Test

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
2024.2493	115.3441	73.77661	156.9115	51.77213	178.9160
2024.2521	166.5986	120.50466	212.6925	96.10402	237.0931
2024.2548	174.6000	128.19214	221.0079	103.62529	245.5748
2024.2575	171.0920	124.63006	217.5539	100.03462	242.1493
2024.2603	168.5686	122.01826	215.1190	97.37599	239.7613
2024.2630	167.9882	121.30274	214.6737	96.58894	239.3875
2024.2658	168.0835	121.24777	214.9192	96.45445	239.7125
2024.2685	168.2004	121.21559	215.1852	96.34335	240.0574
2024.2712	168.2363	121.10519	215.3673	96.15551	240.3170
2024.2740	168.2356	120.95958	215.5115	95.93318	240.5379

Fig.51 Confidence Intervals

Ljung-Box Test:

The Ljung-Box test checks if there is significant autocorrelation left in the residuals of the model. In this case, the test statistic (Q* = 2587.8) is quite large, and the p-value (< 2.2e-16) is extremely small.

This suggests that there is still some autocorrelation in the residuals, which could indicate that the SARIMA model may not have captured all of the patterns in the data. However, this could be a result of high variability in the dataset. Despite this result, the model still appears to perform reasonably well based on the other diagnostic tests.

Forecast and Confidence Intervals: The forecast table provides predicted values for "Arrival to Seen" waiting times along with 80% and 95% confidence intervals. For example, the predicted value for 2024.2521 is approximately 166.59 minutes, with the 95% confidence interval ranging between 96.1 minutes and 237.1 minutes. The confidence intervals widen as the forecast horizon increases, which is expected in time series forecasting due to increased uncertainty over time. These forecasted values can provide the hospital with estimates of future waiting times, helping in staff and resource allocation decisions in the emergency department.

Results for Model-5 Random Forest Model:

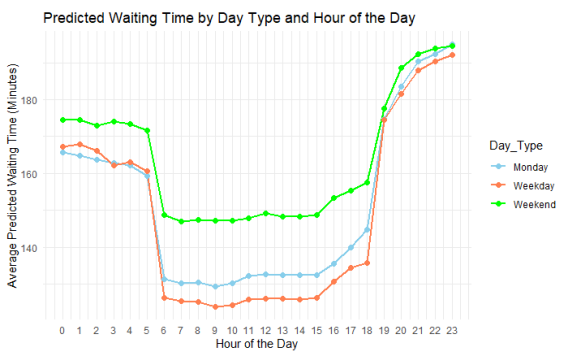

Fig.52 Waiting Time Predictions by Hour of the day – Mondays Vs Weekdays Vs Weekends

Predicted Waiting Time by Day Type and Hour of the Day :

Mondays (Light Blue Line): The predicted waiting times on Mondays show a distinct pattern. The waiting times peak during the early morning hours (0:00 - 3:00 AM) and during the late evening

(starting at 7:00 PM to midnight). The lowest predicted waiting time is observed between 9:00 AM and 4:00 PM.

Weekdays (Orange Line): For weekdays (Tuesday to Friday), the waiting times follow a similar pattern as Mondays but are generally lower during the early morning. The waiting times are at their lowest between 7:00 AM and 4:00 PM and then start to rise, peaking again in the late evening.

Weekends (Green Line): Waiting times are consistently higher on weekends compared to both Mondays and weekdays. The variation during the day is smaller, with the least waiting time occurring between 9:00 AM and 5:00 PM. However, there is a steady increase in waiting time after 5:00 PM, peaking between 9:00 PM and midnight.

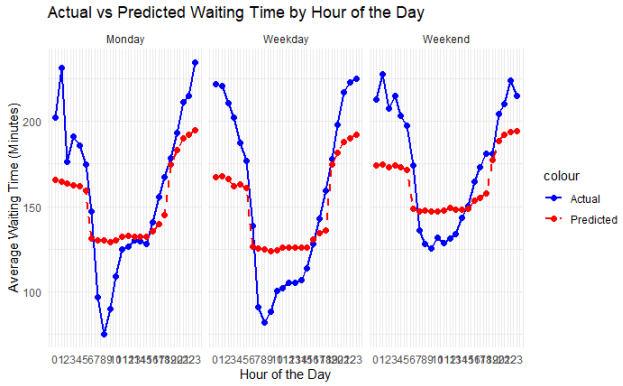

Fig.52 Actual Vs Predictions

Actual vs Predicted Waiting Time by Hour of the Day :

This plot has x-axis is with Hour of the day and the Y-axis is Average waiting time in Minutes. Blue line showing the actual waiting time and Red line showing the predicted waiting time.

Mondays:

Actual Wating times here show a peak during the early morning hours around 2-3AM reaching around 200 minutes and then a dramatic decrease seen throughout the morning and then continue to be low around 7AM to 5PM. Then after 5PM the waiting times start increaing around late evening and continues till early mornings.

On the other hand when you look at the Predicted waiting times we could see a less sharp peak during the early mornings compared to predicted ones and throughout the day the waiting times stay lower than the actual times consistently. Then after 5PM as we know that the waiting times are really high in the actual but which the predictions failed to capture here. This indicates that the model is under estimating the waiting times during the busy times but perfoming reasonably good during the other time of the day.

Weekdays: Then when we look at the Weekday Waiting times there is a similar pattern here as Mondays with a peak around 2-3AM followed by a significant drop during the day time. Then during the late evening the waiting times again rise up but slightly lesser than Mondays.

On the other hand when we look at the predicted waiting times on the weekdays we could see a smoother predictions for mornings and late evenings which again shows that the model is again understimating the times. The flat line in the predicted waiting times indicates us that the model is not cabale of capturing the fluctuations during the day as the actual data shows.

Weekends: For weekends, the actual and predicted waiting times show similar trends. However, the model consistently underestimates the actual waiting time across most hours of the day, particularly during peak hours which is early morning and late evening.

The random forest model captures the general waiting time trends across different days and hours but tends to underestimate waiting times during peak hours, especially during early mornings and late evenings though it performs well throughout the day. But the model handles the weekend waiting times slightly better due to less fluctuations than the other days. While the model performs well for low-traffic periods, further refinement may be needed to improve accuracy during high-traffic times.

Chapter 6

6.1 Discussion:

The Exploratory data analysis helps us to understand the patterns in arrivals, breaches, admissions, and waiting times of the patients in the Emergency department. We could see the peaks in the arrivals of the patients between 10AM to 1PM, breaches highly observed during the late afternoons to evening timings. We also were able to get some key variables like age, admission status, presenting complaints, ED Area types which is an important aspect contributing to predict the patient flow on different days of the week which could help in allocating the resources, plan the staffing and other strategies to treat the patients efficiently.

Pre and Post covid analysis revealed that the intial stages of the didn't effect the volume much for very long. At the beginning of the pandemic which is in 2020 the patient volumes show a decline as we know that the people didn't go to the hospitals during that period but later the flow of patients gradually stabilised returning to the normal levels. The statistical tests performed also proved that there is no significant changes during both the periods indicating us that the hospitals or emergency departments has successfully adpated to the changing circumstances by managing it all with a very efficient operational strategies in he hospital.

From all the highlighted points in the above analysis, we could see peaks in patient arrivals, particularly during late mornings and early afternoons, suggesting that resource allocation, including staffing and equipment, needs to be optimized during these peak hours to prevent breaches in waiting times. Moreover, the pandemic highlighted the importance of maintaining flexible operational capacities, as the ED saw an initial decline in patient volumes followed by a recovery. This demonstrates that the hospital must be prepared to handle fluctuating demand due to unforeseen disruptions like public health crises. On the other hand, the text analysis showed us the issue that were most common but not serious to get admitted, for which hospital must make sure that the treatment is done quick and efficiently so that it could save both resources and time for the staff to handle more critical cases which might really need to be admitted. Therefore, more tailored strategies would help in planning the things efficiently and improve the performance of the ED by reducing the witing times.

6.1.0.Comparing the Models used for Predicting the Daily Arrivals:

Poisson regression model and the negative binomial model has given us the patient count around 323 patients per day, though the negative binomial model performed comparitively well by handling the over dispertion issue which the poisson model was lagging in due to which negative binomial was proved comparitively better performing on the fluctuating data.

However,I have implemented an other model which is prophet model looking at the seasonality and trend components of the data which outperformed both the previous model by accurately capturing all the seasonal and trend combonents of the data. It also provided us the prediction included with lower and upper bounds.

The hospital should adopt the **Prophet model** for forecasting patient arrivals as the seasonal and trend components means that the daily,monthly, yearly and holiday patterns of the data were also efficiently managed by this model providing the upper and lower bounds makes it robust, flexible and best model for this scenario in the hospital to proactively allocate the resources, optimize the staff schedules and prepare for the peak times by ultimately providing efficient serivices to the patients in ED.

6.1.1.Discussing the Models used for Predicting the Admission likelihood of the patients coming in daily:

The results of the logistic regression model for predicting admission likelihood provide critical insights into how different emergency department (ED) areas contribute to patient admissions. The model revealed that patients presenting in "Majors" and "Resus" areas have a significantly higher likelihood of being admitted compared to those in less critical areas such as "Minors" and "Paediatrics." This indicates that the cases,with severe or life-threatening conditions naturally demand admission which leads to requirment of more resources for inpatient care. So, it is very important for the hospital management to have enough reources like staff and beds always available specially during the peak hours for the patients coming in under “Resus” category as they are highly likely to admitted always. Moreover, the small but consistent increase in admission likelihood with rising patient counts suggests that resource allocation must be flexible to accommodate fluctuations in demand.

This also highlights the importance of predicting the patient admission volumes based on arrival patterns. By using predictive models like the Poisson regression to forecast patient arrivals and the logistic regression to predict admission likelihood, hospitals can manage their operational workflow efficiently as they will get an idea on how the patient flow is going to be on different days and hours of the day from these results by avoiding the overcrowding and lack of resource issues with strategic planning. Additionally, understanding that "Minors" and "Paediatrics" areas have lower admission rates helps the management to also focus on efficient outpatient care along with the critical ones, by allocating doctors were there is requirement for same day treatmets. This targeted approach of resource distribution can lead to efficient services to the high patient flow by reduced waiting times, and enhance the overall patient satisfaction in the emergency departments.

6.1.2.Impact of Increased Post-COVID Admissions:

An important observation was made regarding the sharp rise in admission percentages following the COVID-19 pandemic. After the pandemic, there has been a notable change in the way EDs handle the patient admissions, with a higher volume of patients being admitted even for relatively minor issues. This change might be because of multiple factors like, patients who delayed or avoided hospital visits are now coming in to get the treatment and because of the delay, situation might have been worsened and they definitely require admission in to the hospital.

This post-COVID increase in admissions has had a significant impact on waiting times in EDs. As more patients are admitted for observation or treatment, EDs are experiencing longer wait times due to resource constraints. The strain on beds, medical staff, and other critical resources has intensified, leading to bottlenecks, particularly in high-acuity areas like "Majors" and "Resus." This admission trend highlights the need for hospital management to adapt its resource planning to account for the long-term effects of the pandemic on patient flow. Increasing bed capacity, improving triage protocols, and enhancing outpatient treatment options for less critical cases can help alleviate the pressure on EDs in the hospitals.

6.1.3.Comparing the Models used for predicting the waiting time in the hospitals:

The waiting time prediction models used in this study are **SARIMA** and **Random Forest** models, both provide crucial insights into the operational dynamics of the emergency department. The **SARIMA** model effectively captured the seasonal and trend patterns in the waiting times by clearly displaying the fluctuations over the years. This model was useful for understanding how time-series elements such as seasonality and daily patient flow trends impact waiting times. However, there were some limitations in this model like handling complex non-linear relationships, such as sudden increases in demand, which are common during emergency department operations.

To address these issues shown in SARIMA, I have decided to implement the **Random Forest** model which outperformed in handling non-linearities and capturing the interactions between different variables, such as the hour of the day, the day type (weekday or weekend), and patient admission status. This model provided us with a more detailed analysis of how the factors are influencing the waiting times, specially the peak hours of the day. Though, we got some effective results while predicting the waiting time on various times of the day, the model still lacked here by under estimating the waiting times during the peak hours of the day like late evenings which is an indication that there is still scope of improvement here with critical scenarios.

From the hospital operations point of view, the findings from these models suggest several actionable insights. The **Random Forest** model's predictions help hospital management understand which specific times of the day is the patient flow and require planning, particularly during late afternoon and evening periods when waiting times typically increase they need allocate enough staff for the patients. By considering the outputs of these predictive models into operational workflows, hospitals can manage patient flow even better by allocating resources more efficiently, and reduce waiting times which is the ultimate goal.

6.2 Limitations:

Limitations are very important to acknowledge in every study that may influence the findings tried to be achieved. The limitations in my study are:

Inaccuracy due to any missing information or records of the patients. Lacking in segregating the complaints and diagonsis variables due to numerous issues described in the records.

This data belongs to just one NHS branch that might not allow us to generalize the results as every emergency department and hospitals are different from each other. Aditionally, the SARIMA and Random Forest models performed well in predicting waiting times, they struggled to accurately capture the nonlinear and sudden spikes during high-demand periods such as weekends or evenings, indicating a need for more robust models in future studies.

Chapter 7

Conclusion:

The study conducted here to analyse the patient flow and operational efficiency in the Emergency department of an NHS trust in UK using exploratory data anaylysis and predictive models to get the forecasts for patient arrivals, admission likelihood and the waiting time predictions on daily basis. The several insights revealed by the models implemented in different areas gave us some significant points to note and plan the operations and improve the efficiency of the hospital. The **Poisson, Negative Binomial, and Prophet models** for predicting patient arrivals, as well as the **SARIMA and Random Forest models** for waiting time prediction, provide a strong foundation for making more informed, data-driven decisions in the ED. These tools allow hospital management to **anticipate demand** and allocate resources more effectively, especially during peak periods.

Then the logistic regression model for admission likelihood demonstrated that the critical which is "Resus" and then the "Major" categories are the ones which are the areas having significantly higher likelihood of admission. This shows that the the hospitals need to ensure that critical care units are well equiped and has enough staff, particularly during peak hours. Moreover, the impact of the **COVID-19 pandemic** on patient flow has apperently effected by increasing the number of admissions in to the hospital than before.This increase in admissions is also contributing to longer waiting times, suggesting that hospitals should enhance triage processes and outpatient care to manage the influx of patients more efficiently.

From an operational perspective, this study highlights the importance of **flexible resource allocation** to manage both high and low patient volumes. The **seasonal patterns** and daily fluctuations in patient arrivals suggest that hospitals should adjust staffing schedules and bed availability to match these trends specially during the night times and Weekends as the wating times are observed to be higher during these times. Furthermore, future research could explore integrating real-time predictive systems into ED operations, allowing for real-time adjustments to staffing and resource deployment based on actual patient flow.

In conclusion, the application of predictive models in this study offers us insights that will help us to plan some actionable strategies to improve ED efficiency, reduce waiting times, enhance patient care and ensure that everything runs smoothly even during the peak hours.

Thoughts and Suggestions from my observations:

From the analysis I have done, these are the some of the observations and suggestions would be useful to include:

Intially, I could see that the number of Non-Breach is more than the Breach which is a good thing that shows that the hospital is managing to full-fill the 4-hour target up to some extent.

Admitted patients are less than Non-Admitted ones.However, this might be changing according to the system or rules followed by the hospital in the admission processes time to time.

Insights gained from the predictions from our models are:

On average, we can expect upto 300-400 patients approximatly in the Emergency department on daily basis.

Admission likelihood prediction shows that the “Resus” and then the “Majors” are the highly likely categories to be admitted.

And the most important part of our analysis which is the waiting time predictions shows us that though the patient flow is very high on Mondays and 11AM to 5PM during peak hours every day, the waiting times are very less and on the other hand Weekends, times like late evenings to early mornings daily when the patient flow is really low the waiting times are really high.

This clearly shows that the times when the patient flow is low, staffing is not properly allocated due to which the patients are tend to wait for longer times.

Hence, the hospital management should consider this point and allocate the resources during the Night times and Weekends adequately or look into this if there is some other reason behind it which I couldn’t capture from the existing data.

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